

Review

Application of Microalgae to Wastewater Bioremediation, with CO₂ Biomitigation, Health Product and Biofuel Development, and Environmental Biomonitoring

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Featured Application: Potential application of the outcomes of the present study is feasible, for wastewater bioremediation with concurrent cost-efficient biofuel production, along with value-added health product and biofuel development, while also reducing the carbon footprint, certainly after implementation of the necessary regulation sandboxes, to promote environmental and human health protection and also job creation and economy strengthening.



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Abstract: In the current study, the cultivation of microalgae on wastewater-based substrates is investigated for an effective natural wastewater treatment that also generates biofuels and value-added products beneficial to human health. Additionally, the health of ecosystems can be evaluated via microalgae. The utilization of microalgae as bioindicators, biofuel producers, and wastewater treatment providers, under the biorefinery concept, is covered in this article. In fact, bioremediation is feasible, and microalgae culture can be used to efficiently process a variety of effluents. Along with wastewater processing and the creation of value-added substances, bioconversion concurrently offers a viable and promising alternative for reducing CO₂ greenhouse gas emissions to contribute to climate change mitigation. The microalgal biorefinery being considered as the third generation is unique in that it addresses all the aforementioned problems, in contrast to lignocellulosic biomass from agricultural waste in second-generation biorefineries and edible crops in first-generation biorefineries. In particular, one of the most promising natural resources for the manufacture of biofuel, including biodiesel, bioethanol, biomethane, and biohydrogen, is found to be microalgae. Furthermore, products of high value, like fatty acid methyl esters, astaxanthin, β-carotene, DHA, and EPA can be made. Hence, microalgal biomass offers a substitute for the development of biofertilizers, bioplastics, pharmaceuticals, cosmetics, animal and aquatic feeds, and human nutrition products, thus promoting human and environmental health.

Keywords: wastewater; bioremediation; microalgae; biomass; CO₂ capture; biorefinery; health products; biofuel; environmental biomonitoring

1. Introduction

The concept of using microalgae as a source of energy was first proposed half a century ago, though the 20th century saw the real advancement of diverse applications. The first

production of monoalgae crops (*Chlorella vulgaris*) was in 1890. Since 1948, endeavors started within the United States, Japan, and Germany in applications of microalgae (*Chlorella*, *Spirulina*, and *Dunaliella salina*), focusing mainly on their utilization in nutrition and then to explore wholesome and therapeutic applications [1]. In the 1980s and 1990s, the U.S. Department of Energy financed the Aquatic Species Program, pointing at the generation of oils and inevitably biofuels from microalgae [2].

The current study thoroughly reviews the application of microalgae culture to efficient wastewater bioprocessing with concurrent sequestration of CO₂ and development of value-added products for human and animal health, as well as of sustainable biofuels, along with microalgae acting as bioindicators for assessing the condition of ecosystems.

Specifically, in this section, Section 1, the biology, cultivation systems, harvesting methods, and processing technologies for microalgal biomass are introduced. Then, in the following sections, Section 2 discusses the microalgae-based wastewater treatment; Section 3, CO₂ capture and utilization for microalgae photosynthesis; Section 4, generation of value-added health products, and, in Section 5, biofuels; Section 6, specific applications of wastewater bioprocessing; Section 7, the microalgae biorefinery concept; and Section 8, the use of microalgae in environmental biomonitoring.

1.1. Microalgae Biology

Microalgae include both prokaryotic and eukaryotic organisms, which can be classified into microplankton, nanoplankton, ultraplankton, and picoplankton, according to their size. Most of them are eukaryotic and their core resembles that of higher plants, while in their organelles there is chloroplast, where photosynthesis takes place. In various microalgae types, chloroplasts differ in shape and in chlorophyll group combinations, affecting the color of the microalgae. Of the approximately 50,000 species that exist in water and terrestrial environments, only 4000 have been identified. Their basic categorization is based on criteria including the presence of a distinct nucleus, the type of photosynthetic pigment, the type of metabolism, the chemical composition of the cell membrane, the presence of flagella, and the mode of reproduction [3]. Microalgae grow in colonies of various forms depending on the species, such as aggregate, capsidic, coccoid, palmeloid, filamentous, and parenchymatous shapes. Cell walls accumulate oils and polysaccharides, while green microalgae are particularly rich in fresh water. The main polysaccharide of these is the short-chain water-soluble *Chrysolaminarin* (beta-1,3-glucans), which is considered a storage polysaccharide and is the most common biopolymer in the world. Because of the photosynthesis process, green algae, which are especially abundant in freshwater produce starch, which serves as the primary component of storage. They are able to produce oils and fats, though. For example, the carotenoid astaxanthin is primarily found in green freshwater microalgae, specifically *Haematococcus pluvialis* [4].

The metabolism of microalgae is almost like that of organisms that undertake photosynthesis. The intake of nutritional elements from the environment through various biochemical and transport processes is very significant. Carbon and nitrogen are the essential elements during photosynthesis. The primary alterations brought about by metabolic processes include changes in protein, volume, density, mass, chlorophyll, RNA, and vitamin concentration [4]. Elements that are essential in the form of trace elements, such as silicon and iron, are also obtained from organic matter for microalgae growth, but can have toxic effects when they are in high enough concentrations [5]. The biochemical composition of microalgae includes four classes of substances: lipids, carbohydrates, proteins, and nucleic acids. The quantities and proportions vary from species to species. The most energy-rich (37.6 kJ/g) is the group of lipids, which are either polar or non-polar, and their ratio varies depending on the species, metabolic phase, and environmental conditions. Polar ones, such as glycolipids and phospholipids, are found in the cell membrane and have a structural character, while non-polar ones such as tri- and diglycerides and liquid carbons have a storage character. They are generally divided into polyunsaturated fatty acids (PUFAs), which are the structural fraction, and monounsaturated fatty acids (MUFAs) and saturated

fatty acids (SAFAs), which are the storage fraction of lipids. The carbohydrates of microalgae, with an energy content of 15.7 kJ/g, which can be fermented to produce bioethanol or methane, are also separated, like lipids, into structural and storage carbohydrates. Structural ones, such as pectin and cellulose, are substances of the cell wall, while storage ones, such as starch, accumulate inside or outside the chloroplast. The metabolism and composition of carbohydrates vary depending on the type of microalgae. Protein synthesis is a complex mechanism. In the first phase, amino acids are synthesized, and then follows the creation of the peptide chain to obtain the primary protein. Monocytes' photosynthetic microalgae are commonly used to produce proteins with an energy content of 16.7 kJ/g, as they require only salinity, CO₂, and light for their growth [3]. Carbon metabolism begins with the incorporation of glucose into algae cells and the addition of the phosphate group to hexose, leading to glucose 6-phosphate (Glu-6-P), which is easily accessible for cell storage, growth, and respiration. Most glucose is converted into oligosaccharides, polysaccharides, etc. In conditions lacking light, however, microalgae cannot metabolize glucose [4].

The development of microalgae for biomass generation depends on a number of factors: a few of these are light, carbon sources, and sources of supplements including nitrates, phosphates, carbohydrates, and components such as manganese, cobalt, zinc, molybdenum, etc. Ideal temperature, pH, blending within the photobioreactor, oxygen expulsion, and CO₂ take-up in comparable extents are also estimated. Light, temperature, nitrogen, and phosphorus are related to the development rate and lipid substance of microalgae. Hence, these parameters ought to be kept up and controlled for successful biomass generation [4].

Sunlight is essential to the development and output of microalgae. Productivity in any cultivation system is largely dependent on solar energy absorption, and microalgae growth is generally best at 200 µmol/m²/s, or about 1/10 of the maximum radiation of a summer day. Microalgae's photosynthetic pigments are in charge of photosynthesis, which involves collecting light and utilizing the energy it absorbs to produce NADPH and ATP before converting CO₂ and water into carbohydrates. The three primary categories of pigments found in microalgae are phycobilins, carotenoids, and chlorophylls. These pigments are in charge of absorbing light into various photosynthesis-related active radiation (PAR) wavelength ranges, which fall between 400 and 700 nm in the visible spectrum. Blue light (450–475 nm) and red light (630–680 nm) are absorbed by chlorophylls, whereas light between 400 and 550 nm is absorbed by carotenoids, which include lutein, fucoxanthin, α- and β-carotenes, and xanthophylls. However, phycobilins are primarily absorbed between 500 and 650 nm. The rate at which light is bound for photosynthesis is determined by the quantity and quality of light present in addition to each microalgae species' capacity to absorb light [6].

Temperature is also considered an important and critical factor for optimization in large-scale outdoor crop systems. Daily changes in temperature may reduce the productivity of microalgae in lipids. There is also a decrease in cell volume with increasing temperature. The optimum growing temperatures generally range between 20 °C and 30 °C. Microalgae growth is affected by changes in light intensity. Many species of microalgae can withstand temperatures as low as 15 °C, with reduced growth rates, and temperatures higher than optimal can lead to death. In general, low night temperatures and low seasonal temperatures lead to a significant reduction in biomass productivity. Adaptability to temperature varies depending on species. For example, freshwater microalgae, *Scenedesmus* and *Chlorella*, can adapt to temperatures in the range of 5–35 °C, while the ideal range is 25–30 °C [4].

Depending on the type of microalgae salinity, nutrients and pH may affect the growth of their biomass. Salinity affects growth, as microalgae have their own systems for adapting to a specific salinity range. Seawater microalgae may resist higher salinity compared to freshwater microalgae. Most species of microalgae have an ideal salinity. Beneath this salinity, sodium chloride (NaCl) and sodium sulfate (NaSO₄) can be included to progress development. In any case, high salinity (>6 g/L) antagonistically influences and hinders

the development rate of microalgae. The development of microalgae may also be crucially influenced by pH.

Important elements such as carbon, oxygen, hydrogen, nitrogen, potassium, and phosphorus, as well as magnesium, calcium, iron, sulfur, and other trace elements, are necessary for the growth of microalgae. They are obtained either from water and air, or from the growing medium. Nitrogen and phosphorus significantly influence biomass and lipid growth [4]. Nitrogen is one of the richest sources of nutrients found in nature, along with carbon, hydrogen, and oxygen, and is also the main contributor to the increase in microalgae biomass. In microalgae, nitrogen and carbon metabolism are linked. Glutamine synthase metabolizes nitrogen sources. This is estimated to have a high affinity for ammonia and can be easily incorporated into cells. The compounds that can be sources of nitrogen are nitrites, nitrates, ammonia, and urea. Nitrogen, in addition to growth, particularly significantly affects the quantitative content of lipids and their qualitative distribution [4]. In the study of Karayannis et al. [7], the change in lipid content in the produced biomass of *Chlorella vulgaris*, relative to the initial concentration of nitrogen in the culture medium, was examined. The experiments were carried out in pilot photobioreactors in a discontinuous greenhouse and were stopped five days after reaching the stationary phase. Low and very high nitrogen concentrations in the growing medium have a significant impact on biomass production and the percentage of lipids contained. The results of the study suggest that lipid levels could be increased if the culture continued after the stationary phase and the photobioreactor worked in semi-continuous mode while adding potassium, phosphorus, and small amounts of nitrogen. In the case of continuous operation, the best approach for increased lipid productivity is to start with increased nitrogen to achieve high growth rates and, when the culture reaches a stationary phase, add macronutrients intermittently, keeping nitrogen low, so that it resembles semi-continuous operating conditions [7]. Phosphorus is essential to the metabolic processes of microalgae, including photosynthesis, energy transfer, and signal transduction. Phosphatase converts organic phosphorus on the cell surface into orthophosphates to be used in this form by cells. When there is an abundance of orthophosphates, they are converted to a polyphosphate form to be able to prolong growth in conditions of deficiency of extracellular phosphorus. Consequently, changes in growth due to a change in the content of phosphorus in the growth medium are not as direct as those caused by temperature and lighting. Phosphorus concentrations greatly affect lipid synthesis [3]. In photoautotrophic microalgae cultures, the source of carbon is CO₂ in the atmosphere, but it is found in low quantities for the needs of satisfactory microalgae growth. In order to address the inadequacy, carbon dioxide needs to be added either from a clean production source or from exhaust gases, thus achieving the satisfaction of the environmental requirement for its capture. The addition of carbon dioxide should be performed with streams of small bubbles to facilitate its uptake by microalgae culture. The range of 5–10% v/v in carbon dioxide ensures satisfactory growth conditions for most microalgae species, while if the culture medium contains phosphate compounds with the addition of carbon dioxide, the pH drops, and this causes their precipitation [5]. Growth for a large group of microalgae is undertaken with sunlight as the energy source and CO₂ as the carbon source. Crops of this species are called photoautotrophies. Another microalgae class that uses glucose or other organic compounds as a source of carbon and/or energy is called heterotroph. The main advantages of these microalgae are that the dependence on the presence of light is reduced, and their increased productivity in biomass and lipids compared to autotrophies. However, the most important are the flexible mix-food crops, where there are the possibilities for photosynthesis, but also for obtaining energy and carbon from organic or inorganic substances. Under mixing conditions, biomass grows faster than the other two [4].

1.2. Microalgae Cultivation Systems

The two main categories of microalgae cultures are the external type (open tanks) and the indoor type (photobioreactors). In both cultivation systems, the light source and its

intensity play a key regulatory role. In external systems, microalgae receive sun light, while in internal systems they receive it from various types of artificial sources, such as LEDs and optical fibers. However, in both cropping systems, photosynthetically active solar radiation (PAR), when over the saturation threshold, can be a factor limiting growth and reducing biomass productivity. As a result, cultivation systems that are outdoor cannot be effectively controlled, unlike indoor ones that are monitored [8].

These are outdoor cultivation systems in ponds, reservoirs, and elongated basins, as the name implies. This system is the earliest and most fundamental technique for mass-cultivating microalgae. To achieve high biomass productivity at low cost for industrial-scale microalgae cultivation, the open basin system was developed. This is a straightforward technology that works well in many different nations, including those in Asia, Mexico, the United States, Italy, Spain, and the Netherlands. Species of *Chlorella* and *Spirulina* are mostly grown in open lake systems across the world. An elongated basin with vertical walls and a flat bottom that is no larger than 0.5 hectares and has a depth of 0.25 to 0.30 m is the typical configuration for microalgae farming. The parameters that must be considered for the construction of these basins are successful mixing of the growing medium, facilities for planting and harvesting food, the entry of carbon dioxide, drainage, flooding, and cleaning. For cultivating microalgae in open systems, a low-cost medium is used, such as various wastes, which provide the necessary nutrients, phosphorus, potassium and nitrogen, and atmospheric CO₂. Consequently, this technology offers the dual advantage of wastewater treatment and microalgae growth at low cost [9]. Outdoor systems can be divided into two categories: natural containers, such as lakes, dams, and lagoons, and artificial lakes or basins. The most commonly used systems include shallow large lakes, reservoirs, circular ponds, and elongated basins, mostly oval in shape [10].

Indoor culture systems are also called closed-type cultures or laboratory systems. Microalgae culture systems are strictly controlled and monitored and are carried out in special reactors, photobioreactors (PBRs). In these reactors, continuous monitoring of important light and temperature parameters is ensured, as well as of possible contamination by parasites and competing species of other microalgae. Oxygen management is the primary drawback of photobioreactors because a proper degassing system is needed. In microalgae culture, photobioreactors are a commonly used system, independent of climate. Polyethylene, solenoid, tubular, vertical, or horizontal column systems, and flat photobioreactors are among the various varieties [11]. In comparison to elongated basins, flat and tubular bioreactors are the most effective and productive system for achieving a higher biomass yield because they shield microalgae from pathogens, infections, predators, and other competing species. Photobioreactors, depending on the method applied, can be discontinuous, continuous, and semi-intermittent [9]. Stirred tanks are another type of photobioreactor, where stirring is performed mechanically using blades. The mixing efficiency is significantly higher compared to the other types, resulting in satisfactory heat and mass transfer. The disadvantage, though, is that mechanical stirring leads to high shear stress in microalgae cells, which is harmful to cells and an expensive method for their operation and maintenance due to the creation of heat by stirring. The biggest disadvantage of photobioreactors of all types is the difficulty in scaling up installations to achieve production for commercial purposes [12]. Escalation is an area that faces problems in general. In the study of Papapolymerou et al. [13], the cultivation of *Chlorella vulgaris* was studied in a 4 m³ open lake pilot photobioreactor in a greenhouse compared to an automatic laboratory reactor of 25 L. The pilot photobioreactor consisted of three circular tanks with a diameter of 2.10 m and a height of 40 cm placed in series inside the greenhouse. The overflow of one can be the power supply of the next, and therefore they can operate in continuous mode, but also individually in discontinuous or semi-continuous modes. In this study, the function was of a discontinuous type.

The overall setup, the maximum aeration attained in the laboratory bioreactor, the variations in artificial LED (white and red) lighting and natural lighting between the laboratory and pilot bioreactors, respectively, and the pH readings during the development

phase are the primary distinctions between the laboratory and pilot bioreactors. During the summer, the laboratory bioreactor's average specific growth rate was higher than the pilot's. Nonetheless, both the laboratory bioreactor and the pilot photobioreactor experienced the same specific growth rates during the exponential phase of cultivation during the summer. The biomass productivity of the pilot photobioreactor varied significantly between the winter and summer operating seasons, reaching 32 mg/L/d and 90 mg/L/d, respectively. Low winter productivity is mainly due to low light intensity, and less to lower temperature.

Table 1 summarizes the advantages and disadvantages of the most common microalgae cultivation systems.

Table 1. Table of advantages and disadvantages of cultivation systems, inspired by Mantzorou and Ververidis [14] and Razzak et al. [5].

Cultivation Systems	Advantages	Disadvantages
Openly	Easy to build and operate Low installation and operation costs Combined operation with simultaneous wastewater management Use of sunlight	Difficulty managing infections Lower biomass productivity than closed systems Higher water needs and increased costs to satisfy them Water losses due to evaporation Harder biomass harvesting Increased cost of drying biomass Dependence on climatic conditions High demands on land Difficulties in mixing to allow light to be at an optimal value for cell growth and to improve gas exchange
Closed	Control of environmental parameters, pH, temperature and light intensity Ideal for monocultures Minimization of infections	Increased demands on bioreactor design Increased operating costs High energy needs High water needs and increased costs to satisfy them Negative effect of hydrodynamic pressure on cells

Moreover, hybrid systems not only get around the drawbacks of open systems, but also the expensive setup and maintenance costs of closed systems. They can be applied to large-scale crops and have a good cost-effectiveness ratio. To produce high-density vaccines in this instance, the microalgae are first cultivated in a photobioreactor. After that, they are moved into an open system, which aids in achieving the best possible biomass production. Furthermore, there is a significant decrease in the risk of infection in open systems. Indeed, microalgae quickly establish themselves as the dominant species in an open system, effectively competing with other microorganisms [5].

Furthermore, alternative microalgae cultivation systems for biomass production are microalgae biofilms, where the microalgae culture is not suspended, but grows on solid surfaces, presenting significant advantages over classical crops:

- Lower water needs. In normal crop systems, about 200 tons of water are needed to produce 1 ton of microalgae biomass, while 17 tons are required using biofilms, of which 4 tons are used to maintain surface moisture at appropriate levels.
- They are a more economical approach, as the supporting surface of the adherent biomass of microalgae is cheap, easily available, and reusable, while providing the advantage of concentrating the biomass in a small area compared to standard systems that have increased manufacturing costs.
- Increased biomass productivity of 30% has been found compared to suspended crop systems, while higher yields have been recorded in wastewater treatment.

- The harvesting process is greatly facilitated, as the removal of biomass from the abrasion support surfaces is easily achieved. Biomembranes can be used in combination with open systems, and, in particular, rotating membranes in long basin systems offer efficient effluent treatment with simultaneous high biomass productivity. Furthermore, biomass has a low water content, so additional dehydration is not required. Colonies that remain on the surface after harvest are excellent vaccines for the next growth cycle.
- Microalgae biofilms achieve, overall, more satisfactory light penetration compared to open culture systems, as although the penetration is almost the same in the upper layer of both systems, in the lower part of open systems the intensity of sunlight is much lower than in biofilm systems.
- When membrane culture systems grow in lakes, lagoons, or oceans, it facilitates the control of their growth due to the high concentration of connected cells in the substrate and their limitation to membranes.

The disadvantages of this technology are:

- The possibility of removing biomass from the solid surface. This can lead to the sedimentation and loss of bound pollutants in waste culture systems and consequently their release into the aquatic environment.
- Microalgae biofilms can cause damage to metal surfaces, such as corrosion and discoloration, through the produced microbial products [14].

Furthermore, the immobilization of microalgae in calcium alginate is an alternative method in which the cells of the microalgae are physically trapped and encapsulated inside the gel globules. The natural alginate polymer, derived from brown marine algae, through the multiple blocks of the guluronic acid chains it contains, and the calcium that acts as a binding agent, forms a solid body, where the immobilization of microalgae takes place. This method facilitates the recovery of biomass, which is performed either by simple sieving or by precipitation. These systems use fluidized or fixed beds, in closed columns, and can successfully treat waste, within a short period of time [15].

1.3. Microalgae Harvesting Methods

Harvesting is the procedure of utilizing different methods to separate the biomass of microalgae from the growing medium. The separation of cells from the sparse suspension of cultures, but also the repeated frequency of the process due to rapid growth, makes the harvesting of microalgae much more expensive than the harvesting of terrestrial plants. The high costs of harvesting are due, in addition to the small size of the cells (3–30 μm), to very sparse crops (lower biomass concentration $<0.6 \text{ g/L}$, which is close to the density of water) [8].

In particular, factors such as the type of microalgae, the density and size of the cells, the specifications of the final product, and the reusability of the culture medium affect the harvesting method. The development of a dilute suspension (0.02–0.05% dry solids), small cell size ($<30 \text{ mm}$), negligible density difference between biomass and the culture medium, and high growth rates requiring frequent harvesting make the microalgae harvesting process a difficult and costly task. The cost of harvesting in open microalgae cultures amounts to 20–30% of the cost of biomass produced, and, together with drying, they reach 90% of the total cost in terms of equipment [16].

As there is no fully effective and competent method of harvesting and dehydration, combinations of methods in two stages of the process are proposed. In many cases, centrifugation and filtration are applied, followed by flocculation to improve harvesting efficiency and reduce costs [16]. In the two-step processes, the dilute microalgae suspension of the primary harvest is concentrated into a slurry with 2–7% TSS (total suspended solids) and then in a second step, dehydrated to 15–25% TSS, requiring more energy to go to drying, where the total solids will reach 90–95%. In the case of biofuel production and in order to achieve low harvest costs, a combination is made with more than two methods. In addition, the choice of harvesting method influences the quality of recycled water and is

critical to avoid wasting water from microalgae cultivation [17]. In Table 2, a comparative evaluation is made of the main methods of harvesting microalgae and the advantages and disadvantages of each, inspired by other works.

Table 2. Table of advantages and disadvantages of harvesting technologies, inspired by Kadir et al. [17], Mantzourou and Ververidis [14], Singh and Patidar [16], Garbowski et al. [11].

Technology	Advantages	Disadvantages
Centrifugation (physical method, where cells are separated based on density and in which the heavier ones move away from the axis while the lighter ones approach it).	It is applied to all kinds of microalgae. It has a high recovery rate. No chemicals are used. It is the fastest and most efficient process.	In large-scale applications and for low-value products such as biofuels, energy needs and investment costs are very high. There is no possibility of easy scaling. Possible cell rupture due to shear and gravity forces.
Flocculation (chemical method, in which the microalgae of the suspension settle as aggregates with the help of thrombotic substances, with which their negative charge is neutralized. In addition, it can be performed with self-flocculation, bio-flocculation, and electro-flocculation.)	Quick and easy technique. It is used in large-scale applications. Less harmful to cells. It is applied to a wide range of microalgae. Less energy requirements. Self-flocculation and bio-flocculation are economical methods. Non-toxic and edible organic substances can be used. It is usually used in combination with filtration, centrifugation, and precipitation.	Cost of chemicals Sensitivity to pH changes (some of the flocculants can cause a pH increase even up to 11). Difficulty separating flocculant from recovered biomass. The effectiveness depends on the thrombotic used. It is difficult to recycle the growing medium. Possibility of infections.
Filtration (physical method of separating microalgae biomass from the culture medium through a filter or porous membrane. The culture goes through filters that operate by gravity, under pressure or vacuum).	High biomass recovery. Economic. No chemicals are required. Low energy consumption (natural filter and pressure filter). Water recovery.	Slow method. Not suitable for small cells. Blockages occur in the pores of the filters and their replacement or cleaning increases operation and maintenance costs. High power consumption (vacuum filter).
Flotation (a physical method of gravitational separation, in which air or gas is supplied and bubbles carry suspended matter to the top of the wet surface from where it is collected. It can be performed by dissolved air (DAF), dispersed air (DiAF), electrolytic flotation and ozone (DOF).	It is used in large-scale applications. Low investment costs and need for small spaces. Short uptimes. No chemicals are required. High efficiency. It is combined with centrifugation and filtration.	Doubtful financial viability due to high operating costs. It is applied to small cells (<500 µm). Use of coagulants. Not applicable to marine farming.

In addition to the above methods, the method of natural sedimentation is used, which is simple and economical, but takes long residence times, is not efficient, presents a risk of biomass spoilage, and produces a low concentration of pulp. Recovery efficiency (RE) and concentration factor (CF) are two metrics that can be used to assess how effective the harvesting process was. Together, RE and CF show how effective the process is at separating the biomass of the recovered microalgae regarding mass and volume. Here are the definitions of RE and CF:

RE = removable/cell mass in the initial culture

CF = concentration of microalgae in the final product/initial concentration of microalgae in culture

The mass and concentration of microalgae can be measured as chlorophyll content, optical density, and dry weight of the culture [16].

The appropriate harvesting method is followed according to culture density and cell size of microalgae, as well as production purpose. Nevertheless, none of them has been chosen as completely economical and suitable for the production of microalgae biomass on a large scale [14]. As the four main applications of microalgae are biofuels production, food and feed, value-added products, and the restoration of water quality during wastewater treatment, there are six main criteria, which lead to the appropriate choices of harvesting techniques, as shown in Table 3.

Table 3. Basic criteria for choosing a harvesting method, inspired by [14].

	Criterion	Description
1	Biomass quantity	When it is required to produce a large amount of biomass; therefore, the application needs a large area.
2	Biomass quality	When the quality of harvested cells should be good and not release organic matter.
3	Cost	When low operating costs are needed, it is the first requirement.
4	Process time	When harvesting should be performed at a fast pace
5	Special species of microalgae	When a method specific to the needs of a particular species of microalgae needs to be applied.
6	Toxicity	When the biomass produced requires as a first requirement the absence of toxicity.

1.4. Microalgal Biomass Processing Technologies

In general, the conversion of biomass into fuels is performed through thermochemical, biochemical, and physicochemical processes (Figure 1). The conversion of microalgae biomass is mainly performed by thermochemical methods, which are classified into five categories: liquefaction, pyrolysis, gasification, carbonization, and combustion. The method chosen each time depends on a series of parameters, with the required final product and its application playing a dominating role.

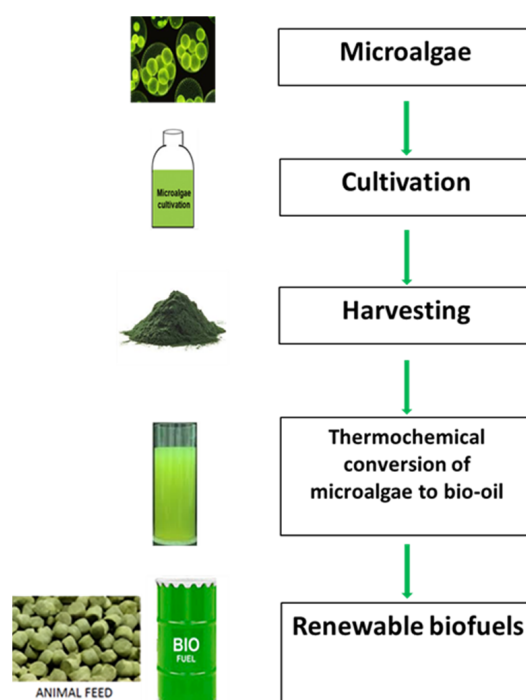


Figure 1. Biofuel production from microalgae.

Hydrothermal liquefaction (HTL) achieves the conversion of microalgae biomass, without a pre-treatment stage for drying, and thus avoids operating and energy costs, which due to the increased water content (~80%) of biomass would be particularly increased. In addition, heat transfer is successfully exploited due to the small size of the microalgae, and this supports the efficiency and cost-effectiveness of the process. During the liquefaction process, the microalgae biomass, in the form of a slurry, enters a reactor, where using an aqueous medium, it is converted into liquid fuel under high pressure conditions (5–20 MPa) and at temperatures from 250 to 370 °C. Usually the biomass of microalgae makes up 5–20% of the total power supply of the reactor, while the rest is the liquid medium. When water is used as a liquid medium, the main product is a mixture of crude bio-oil that is more deoxygenated and degraded than that produced by the pyrolysis method. It can be refined to be used in a variety of ways because it is similar to crude mineral oil. The process residue is bitumen, a mixture of cyclic and polycyclic high-boiling hydrocarbons (such as 2–5-ring aromatics). The conditions applied to liquefaction may change and affect the final products produced, depending on their end use. Below 250 °C, hydrothermal carbonization takes place, as hydrochar (carbon residues), a material similar to coal (coal), is produced as the main product, while when the temperature rises above 380 °C, the process is referred to as hydrothermal gasification and its main product is syngas. The advantage of gasification over liquefaction is that the syngas produced is a mixture of hydrogen, carbon monoxide, carbon dioxide, and methane, which does not contain nitrogen and has a high energy content [9].

A major disadvantage of liquefaction is the large amount of organic material remaining in the process fluid, resulting in loss of energy charge and thus hindering the cost of the process. This liquid contains from 8.01 to 21.55% of the carbon of microalgae, in the form of acetic acid and acetamide, and the difference in distribution is due to the type of microalgae. In addition, when it is discharged without treatment into the environment, it creates an environmental burden. Therefore, efficient management or reuse of the liquid medium is an important stage for the economic viability of the process.

By gasifying biomass with supercritical water (SCWG), residual organics can be converted into hydrogen and methane and easily separated from water. Supercritical or superheated water is water of 374 °C at a pressure of 218 atm, where a liquid homogenized phase is formed that has gaseous and liquid properties. With this method, 99.2% of the carbonate loads of liquefaction water have been recorded, as C-C bonds are broken down, allowing its reuse in microalgae cultures. From the complete gasification achieved, the hydrogen generated can be applied to the liquefaction of microalgae biomass to produce crude bio-oil. After being treated with supercritical water, minerals recovered from liquefaction water can be utilized as fertilizers for plants or other biomass sources. Lastly, from an energy standpoint, compared to HTL alone, the combination of HTL and SCWG can increase energy recovery from microalgae biomass by 5.53 to 18.30% [18].

Pyrolysis achieves decomposition of microalgae biomass under temperature conditions (400–600 °C) and in an inert environment with main products, bio-oil, solid biochar, and gases such as synthetic gas. The produced bio-oil is more stable than that of liquefaction and its quality depends both on the operating parameters of the pyrolytic process and on the type of microalgae. Pyrolysis can be carried out at slow rates of temperature increase (5–10 °C/min) and last an hour, so it is referred to as slow pyrolysis, while when carried out at 600 °C for a few seconds, it is defined as rapid pyrolysis, and its application is more widespread, especially in continuous processes [9]. Catalysts contribute to increasing the efficiency of the pyrolysis process and shortening its timing. Catalytic cracking can increase product selectivity or reduce the effect of reaction conditions, and can be applied either on-site or off-site. For on-site catalytic pyrolysis, the raw material and catalyst are mechanically mixed and then fed simultaneously into the pyrolysis reactor. For catalytic pyrolysis outside, the products, immediately after the reactor, pass through beds, which contain the catalyst. Because pyrolysis generally requires the moisture content of the mate-

rial not to exceed 10% wt., microalgae biomass should undergo dehumidification, which can be achieved either by dissolved air flotation or centrifugation [19].

Gasification is a process applied at temperatures above 700 °C and under controlled oxygen conditions. It is mainly applied to low-moisture biomasses, but can also be used for microalgae biomass. It can be either conventional or supercritical gasification. Conventional gasification is oxidized with air, oxygen, or steam in a gasifier. The temperature in the gasifier is between 700 and 1000 °C and the process steps include drying of the remaining moisture, pyrolysis to break down the solid mass into smaller molecule structures, oxidation of biomass, and finally gasification of pyrolysis products. The products produced by the gasification of microalgae are synthetic gas, char, tar, and ashes. As mentioned above, supercritical gasification is more efficient for microalgae biomass than the conventional method due to its increased humidity. It is applied under 400–500 °C and 24–36 MPa, with products such as those in the conventional method [9].

Hydrothermal carbonization of microalgae biomass is a mild process carried out between 150 and 250 °C at a pressure below 100 bar with long lead times, usually over an hour. The main product of the process is a solid substance, biochar. This contains essential nutrients and minerals, and can be used as fertilizer in agricultural crops and many other applications. Experimental yields of 60% weight in less than an hour at 200 °C at a pressure of 20 bar have been recorded. The carbon content derived from carbonization varies between 20 and 60%, and the exact percentage depends on the type of microalgae and operating conditions. In addition, the char produced absorbs the hydrophobic lipids present in the biomass mixture, which are recovered using solvents [9].

The combustion process is carried out at temperatures around 850 °C, in the presence of oxygen in boilers or furnaces, where microalgae biomass enters with a humidity of more than 50%. The air supply is in excess by 10% of the burner supply, to ensure heat release and complete reactions. Many times, and in order to reach the ideal percentage of 20%, biomass needs to be pre-treated when it has high humidity. During combustion, the energy stored in the biomass of microalgae, due to photosynthesis, is converted into hot gases. The heat generated by the combustion process can be converted by turbines into electricity. In general, because the stage of drying as pre-treatment is costly, low-energy drying is preferred as a technologically advanced process, in order to make the combustion of microalgae biomass more economical overall. Alternatively, the simultaneous combustion of microalgae with other raw materials such as natural gas and coal is promoted, so as to reduce greenhouse gas emissions of the combustion mixture, as microalgae during their cultivation phase absorb the CO₂ produced by combustion [9].

2. Microalgae-Based Wastewater Treatment

2.1. Bioprocessing of Wastewater

Microalgae, beyond acting as biological indicators, are also used for the treatment of various wastewater streams, as they can metabolize organic compounds, heavy metals, nitrogen, and phosphorous found in wastewater treatment plants to increase their biomass, thus solving the problem of eutrophication phenomena in the final recipients. Zabochnicka [20] studied ammonium nitrogen from industrial wastewater utilizing immobilized microalgae, which reached an expulsion proficiency of up to 60% (*C. vulgaris*) and 42% (*S. armatus*). An effective means of wastewater treatment with satisfactory results is high-performance tanks (HRAPs), which are open culture systems, but also immobilized cell methods. The biomass produced by these systems can be utilized for animal feed, and for the production of biofuels and other products, but not as food. Indicatively, the removal of 99% of phosphates from the effluents of the anaerobic digester of a starch industry plant with *Spirulina platensis*, cultivated in HRAP, and the removal of pigments from textile industry effluents from the microalgae *Chlorella vulgaris* via the method of immobilization in alginate beds, have been recorded.

The use of microalgae can also be undertaken with a combination of different species, with satisfactory results, such as the successful withdrawal of nitrogen and phosphorus

from domestic wastewater operating in semi-continuous operation via the use of immobilized *Chlorella vulgaris* together with *Scenedesmus obliquus* and other microorganisms. In this case, oxygen produced by the photosynthetic mechanisms of microalgae reduces the aeration needs of wastewater, and is used by aerobic microorganisms for the degradation of organic pollutants, but also contributes to the removal of volatile substances produced during anaerobic treatment.

As biological markers, *Pseudo kirchneriella subcapitata*, *Dunaliella tertiolecta*, *Isochrysis galbana*, and other species of the genus *Chlorella* have been used for many years. In relation to the assessment of phosphorus and nitrogen presence in freshwater zones, *Ankistrodesmus convolutus*, *Chlorella vulgaris*, and *Scenedesmus quadricauda*, which are of tropical origin, are widely used [21]. The main approaches to processing and removing nutrients from different types of wastewater in large-scale experiments are the use of open systems and bioreactors. Efficacy varies depending on the type of microalgae used for each effluent and depending on the cultivation system followed, but it is proven that in open systems, nitrogen and phosphorus removal yields are more satisfactory than those of bioreactors [22].

2.2. Large-Scale Wastewater Treatment

The creative idea of building the first large-scale plant, where wastewater is treated with microalgae and simultaneously produces bioplastics, biomethane, biofertilizers, and clean water for irrigation, was developed within the framework of the EU (funding from the EU program INCOVER). The research and larger-scale activities took place in Germany and Spain over a three-year period. The *Agròpolis* plant, situated in the Barcelona metropolitan area, mainly used a mixed microalgae and cyanobacteria culture that developed in three horizontal hybrid tubular photobioreactors to treat a mixture of agricultural waste and urban wastewater. Heterotrophic bacteria grow in culture at the same time as other bacteria, but at slower rates because the inputs contain less organic matter. Following the biomass separation process, the treated water is put through three adsorption columns for nutrient recovery, ultrafiltration, and disinfection before feeding energy crops (rapeseed) with a smart irrigation system. Anaerobic co-treatment (AcoD) of the generated biomass with secondary sludge for biogas production is carried out concurrently. The biomass is subsequently enriched in an absorption column using mixed liquid from the photobioreactors. To create biofertilizers, the digestate from the anaerobic reactor is additionally processed in wetlands for sludge treatment. Utilizing microalgae and bacteria mixed cultures grown in municipal wastewater, the three photobioreactors, each with a useful volume of 11.7 m³, were vaccinated. An inclined plate separator was placed after the photobioreactors to split up the biomass and clean the treated water. The water kept in the three storage tanks is pumped through the separator, which has two chambers. In the first chamber, the water is mixed with a coagulation liquid (polyamide chloride, with 9% aluminum). The water travels through a second chamber after passing through the mixing chamber, where the solids sink to the bottom and the water remains in the upper section of the separator. The homogenization tank receives 6.4 m³ of agricultural waste and 0.5 m³ of urban wastewater (which was previously treated in a ventilated septic tank) every night. These materials are mixed there to provide the photobioreactors with an input that has the right amount of nutrients for cyanobacteria growth. This operation is performed early in the morning so that nutrients are ready for use for biomass growth in the daytime. Biomass is produced during the microalgae treatment process of wastewater, primarily due to the nutrients available (primarily nitrogen and phosphorus) in the inputs and solar radiation. Microalgae simultaneously produce the oxygen required for bacteria to aerobically break down organic materials in wastewater through photosynthesis. Thus, among the basic benefits of photobioreactors are microalgae biomass production and wastewater treatment without the need of exterior ventilation. The concentrations of nutrients in the effluents used in the photobioreactor input and completely removed were phosphates <2 mg/L, ammonia <10 mg/L, and nitrites and nitrates <15 mg/L. The microalgae species that developed in the photobioreactors were observed under a microscope and were the diatom

Cyclotella sp., the green algae *Oocystis* sp., and the cyanobacteria *Synechocystis* sp. To attain a nutrient concentration that is appropriate for cyanobacteria, a blend of agricultural and municipal effluents is fed into each of the three photobioreactors. Because cyanobacteria are prokaryotic photosynthetic microorganisms that can digest and store glycogen, polyphosphate, and polyhydroxyalkanoates (PHAs), with polyhydroxybutyric acid (PHB) predominant there is interest in their growth. The use of wastewater in order to gather cyanobacteria from a cultured mixed with various microalgae is an innovation, rather than using only genetically modified crops that require costly processes. Once cyanobacteria are selected in photobioreactors, they undergo nutrient stress in laboratory experiments to select the best conditions to promote PBB accumulation.

The biomass obtained is subjected to thermal pre-treatment and then used as a substrate in an AcoD process, with secondary sludge treatment from municipal wastewater treatment plants. A thickener increases the concentration of solids of the separated biomass from 3–8 to 30 g/L. Then, the biomass is taken to the AcoD plant for biogas production. The biogas produced is stored before the upgrading process, which aims to directly use it as a biofuel by increasing the content of CH₄ from 60–70% to 99%. In the upgrade process, the mixed liquid of the photobioreactors is used as an absorption solution for the simultaneous removal of CO₂, H₂S, NH₃, and volatile organic chemicals, with the advantage of reducing energy costs. The anaerobic digester's sludge is a nutrient-rich byproduct. Previous studies supported the effectiveness for dehydration and stabilization of sludge treatment of INCOVER plant's digestate in a sludge treatment wetland (6 m²). A small plot of land is irrigated with the recovered water (approximately 250 m²). The crop chosen is rapeseed (*Brassica napus*), an autumn–winter cycle crop used for mustard production and biodiesel production. The project is ongoing (at the publication stage of this study, November 2018), and so far, biomass production has been recorded amounting to 2.2 kg volatile suspended solids (VSS)/d while the performance in effluent treatment has been considered completely satisfactory. The authors also expect satisfactory results for the production of bioplastics, biomethane, and biofertilizers. This plant presents, at the production process level, a sustainable approach to wastewater management, where waste is converted into resources for the production of value-added products and biofuels [23].

2.3. Removal of Medicinal/Pharmaceutical Substances

The increase in global water pollution with a plethora of pharmaceutical substances (PCs), such as antibiotics, painkillers, steroids, antidepressants, and antipyretics, used on a daily basis by the ever-growing world population, is nowadays a growing environmental issue due to their significant ecotoxicity and the resulting health consequences. These are complex chemicals that are transported by air, water, and soil, resulting through the food chain to accumulate in living beings and humans, due to their hydrophobic and important resistant property. Pharmaceuticals from urban wastewater have been demonstrated to be extracted by microalgae in both natural and experimental conditions [24]. Existing technologies for their removal in wastewater treatment systems are advanced oxidation processes (AOPs), absorption by activated carbon, biodegradation by bacteria, and fungi and enzyme processing. Increased cost, unsatisfactory efficacy, and production of process byproducts with a higher burden than the main active substances of drugs (APIs), enhance the search for new methods to remove these substances [25].

Using microalgae for the bioremediation of effluents contaminated from PCs attracts particular scientific interest, as their characteristics, such as their photosynthetic action by exploiting sunlight and capturing CO₂, easily manageable quantities in processing streams, and fast growth, are combined with simultaneous biomass production, leading to biofuels, soil improvers, and high-value products. In particular, mixed crops with their metabolic potential have the required flexibility to survive and thrive in extreme environments. The mechanisms of removal and metabolism of PCs by microalgae are bio-adsorption, bioaccumulation, and intra- and external cellular biodegradation.

Bio-adsorption is an extracellular process that depends on the hydrophobic structure and active molecules of PCs and on the type of microalgae. The cell walls of microalgae have polymer lattices that resemble cellulose, pectins, hemicelluloses, proteins, and lignin, while they are negatively charged due to the presence of carboxylates, phosphates, and amines. For example, adsorption by various microalgae ranging from 0 to 16.7% of active substances, such as ibuprofen, metoprolol, diclofenac, and paracetamol, has been recorded, while progesterone and norgestrel cells of *Scenedesmus obliquus* and *Chlorella pyrenoidosa* have been found [25].

Bioaccumulation is a metabolic process in which microalgae absorb organic pollutants along with the nutrients they need for their growth. For example, *Desmodesmus subspicatus* accumulates about 23% of radiolabeled 17α -ethylenestradiol (14C-EE2) in 24 h, while *Chlamydomonas mexicana* and *Scenedesmus obliquus* accumulate carbamazepine [25].

Biodegradation is the most successful way for microalgae to minimize the organic load, as they form simpler molecules through catalytic degradation of the complex substances of their environment. An example of intracellular biodegradation is the action of *S. obliquus* and *C. pyrenoidosa*, where, in an aqueous medium, they biodegraded 95% of the progesterone content. Biodegradation of 30–80% of ibuprofen, caffeine, and carbamazepine from microalgae in municipal wastewater has also been recorded [25].

Extracellular polymers (EPs) produced by microalgae and diffused into the environment are polysaccharides, proteins, and enzymes. These can create, through their interactions, a film that acts as an external mechanism of degradation of organic substances in the environment, such as PCs, whether they are dissolved in the medium or are in solid or colloidal form. With the accumulation of nutrients from negatively charged polysaccharides and proteins, xenobiotic substances are also absorbed, resulting in the neutralization of toxic substances present in the growth medium of microalgae, while at the same time internal cell degradation is affected through the produced byproducts [25].

The efficiency of the removal of active pharmaceutical ingredients (APIs) depends on the type of microalgae, culture method, operational parameters, and physicochemical characteristics of the active substance, and varies from zero to complete removal [26].

In the study of Gojkovic et al. [26], eight green microalgae were selected from the collection of local (Swedish) microalgae grown in municipal waste, which showed excellent performance in nutrient removal and biomass production, and were compared to *S. obliquus* RISE in their ability to remove APIs. The vaccines were prepared in 100 mL conical bottles with Bold's Basal Medium as the growth medium, and the time it took to develop them was from 7 to 9 days. Each crop was then inoculated in a photobioreactor with an initial biomass concentration of about 100 mg/L and an artificial growth medium. Each experiment lasted 12 days in the presence and absence of APIs. The photobioreactor used for the experiments was laboratory-constructed, with a flat usable volume of 1.2 L, illumination by LED lamps in the pattern of day and night every 12 h, air supply with 3% CO₂, and temperature regulation at 20 °C and pH at 7.2 ± 0.5 . The experiments were conducted on 19 APIs and indicatively showed that water-soluble active substances, such as caffeine, carbamazepine, oxazepam, tramadol, fluconazole, codeine, and trimetoprim, were persistent, as 60% of their initial amounts remained after 12 days in the microalgae cultures, while the strongly lipophilic Biperiden, Trihexyphenidyl, Clomipramine, and Amitriptyline accumulated in the biomass (88%), especially in the case of microalgae that showed increased biomass productivity. In particular, *Chlorella vulgaris* 13-1 and *Chlorella saccharophila* RNY had high biomass production (2.6 and 5.4 g/L on a dry basis, respectively) and the best removal yields for all 19 APIs after 12 days.

It was found that the removal of APIs is strongly dependent on the microalgae species, while the total lipid content ranged from 13 to 30% on dry biomass, which was also dependent on each species. Lipophilic substances were removed from the culture medium quickly and efficiently but accumulated in biomass more than polar substances. Only two of the eight microalgae were unable to respond satisfactorily, while all others were equal to or better in terms of efficacy than *Scenedesmus obliquus* RISE, the reference strain [26].

3. CO₂ Capture and Utilization (CCU)

Renewable and green energy sources, such as solar, wind, hydrodynamics, and biomass, are the most effective in minimizing CO₂ emissions by reducing their carbon footprint. In particular, biological methods are considered to be dynamic and promising for efficient CO₂ capture and storage. Microalgae absorbing CO₂ from power plant exhausts can be used to lower greenhouse emissions into the air. In relation to CO₂ emitted, using microalgae for the production of biofuels could, through the photosynthetic process, balance the production of CO₂ by power plants, resulting in the net release of CO₂ tending to zero [21].

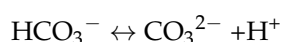
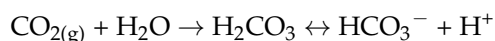
In particular, microalgae are more effective through photosynthetic mechanisms against terrestrial plants. Microalgae have faster growth rates and efficiency in capturing CO₂ (10–50 times more) than terrestrial plants and can use higher concentrations of CO₂, as emitted by the exhaust gases of power plants and other sources. Microalgae can absorb CO₂ from the atmosphere, power plants, and soluble carbonates such as Na₂CO₃ and NaHCO₃, and cope with extreme conditions. In relation to the exhaust gases of power plants, the use of microalgae must be preceded by cleaning from SO_x and NO_x, as these have a negative effect on biomass productivity due to the reduction in the pH of the crop, leading to reduced carbon sequestration and reduced growth. An additional limiting factor is the high temperature of the exhaust gases, which makes the use of heat-resistant microalgae strains mandatory [27]. For example, the marine microalgae *Chlorococcum littorale* can grow at CO₂ concentrations of up to 40%, while the same resistance (40%) has been shown by strains of the genus *Chlorella* from hot springs, and even at a temperature of 42 °C [21].

CO₂ is one of the controlling factors for microalgae photosynthesis. Microalgae capture CO₂ under alkaline conditions and intensify their biomass. Increasing CO₂ levels improves photosynthetic efficiency, which leads to higher biomass yield. Below pH 5, the capture of CO₂, which is converted into biomass with a simultaneous release of oxygen, is not facilitated. Additionally, the permeability of cells and the forms of inorganic salts, and consequently the reactions between, them may be impacted by pH. Carbon binding takes place in chloroplasts, and, specifically, the process of carbon binding CO₂ from the microalgae takes place during the Calvin cycle and is called the dark phase of photosynthesis. Photosynthesis involves two stages: the light phase, where reactions occur only when cells receive light, and the dark phase, during which carbon capture reactions take place, both in the presence and absence of light. The enzyme carboxylase/oxygenase of ribulose 1,5-diphosphate (Rubisco) produces 3-phosphoglycerate, which is the first product in the series of carbon capture reactions. Taking into account the average chemical composition of microalgae biomass, approximately 1.8 kg CO₂ is required to produce 1 kg of biomass. Most microalgae species develop well in CO₂ concentrations from 0.038 to 10%; however, some species demonstrate high resistance to higher concentrations. Some microalgae grow well even at higher CO₂ concentrations, up to 100%, while biomass productivity and carbon sequestration are usually higher at low concentrations. The reason for the decrease in growth in large CO₂ concentrations for most species is that the pH from the production of carbonate ions decreases, resulting in the extension of the latency phase, and consequently lowering growth rates in the exponential phase.

The most important thing about microalgae application for the CO₂ capture process is that biomass is produced at the same time, which has great potential to be changed to numerous bioproducts, such as biofuels, colors, cosmetics, food, feed, and other products. Therefore, microalgae are a satisfactory and sustainable choice for the scientific community as an important CO₂ capture agent at global level. A critical prerequisite is the development of an effective microalgae cultivation system, since changes in physicochemical parameters play an important role in enhancing biomass productivity and CO₂ capture [28].

Photobioreactors utilize CO₂ far more efficiently when compared to open lakes, where the majority of the CO₂ is released into the atmosphere. In large-scale applications, design and operation are very important, so as to achieve good stirring and aeration in order to facilitate light absorption and CO₂ capture [13]. Before being fed into photobioreactors

(PBRs), CO₂ gas from point emission sources is compressed, cooled, and filtered for efficient CO₂ capture. Mass transfer is a very intricate process. The system consists of three phases: mass transfer from gas to liquid, gas to solid, and liquid to solid. In this system, CO₂ is the gas, the culture medium is the liquid, and the microalgae cells are the solid. In order for the bubbles to rise and disintegrate, CO₂ is blasted through the aeration device to the PBR's bottom. When CO₂ gas first reacts with water, carbonic acid (H₂CO₃) is created. This acid quickly decays to produce bicarbonate (HCO₃[−]) and the carbonate radical CO₃^{2−}. As the following reactions demonstrate, cells primarily take up carbon in these two forms rather than directly absorbing CO₂, which happens at a much slower rate:



The conventional aeration system technique for adding CO₂ results in uneven bubbles and does not ensure uniform mass transfer. This disadvantage is removed by the use of hollow fiber membranes. Thus, significantly less CO₂ is used and the crop growth rate is optimized. Aeration is improved by the technique of using diffuse fine bubbles to supply the CO₂ current, in microalgae culture, as it increases mass transfer to the crop, avoids CO₂ loss, and is considered a lower-energy-cost technique. While in conventional ventilation systems the size of bubbles ranges from 1 to 3 mm, in microbubble systems the range is 200–800 µm. Due to the large surface area to volume, they have the possibility of long residence time in the suspension of the culture, as they move slowly towards the top of the bioreactor, and thus the transfer of mass between gas and liquid takes place at higher rates. By applying such a system to an airlift bioreactor, an increase of 20–40% in the specific growth rate of microalgae culture and 6–8 times higher chlorophyll content were recorded. By optimizing physicochemical parameters such as light, temperature, and pH, microalgae cultures achieve greater CO₂ sequestration, but increase the total cost of cultivation. In this context, it is necessary to improve the resilience of microalgae through genetic engineering. Using microalgae models such as *Chlamydomonas reinhardtii*, genetic manipulations are performed to enhance CO₂ fixation, targeting carbonic anhydrase and Rubisco genes, or by interfering with metabolic pathways that lead to enhanced lipid and carbohydrate production [27].

4. Concurrent Development of Value-Added Products Beneficial to Human and Animal Health

Microalgae produce several products for use as raw materials for food, cosmetics, animal feed, fish feed, biofuels, and other value-added products. Many types of antioxidants, carotenoids, polymeric enzymes, lipids, natural dyes, polyunsaturated fatty acids, peptides, toxins, and sterols can be produced from different types of microalgae, which are used in a range of industrial products. Natural dyes are also produced for use in the pharmaceutical industry. Microalgae also produce acetyl acids, beta-carotene, agar, agarose, alginates, and carrageenans, which are used as viscosity-modifying agents in medicines and foods, as well as polyunsaturated fatty acids and lutein, which are used to treat degenerative diseases.

In particular, marine microalgae may produce polyunsaturated fatty acids (PUFAs), which are useful in disease treatment and/or production of food. Large volumes of oils and fats containing long-chain omega-3 and omega-6 fatty acids (LC-PUFAs) can be produced by a number of microalgae species. Human nutrition and health, especially for the mental and physical development of infants, depend on LC-PUFAs.

Microalgae create sterols, which help in cardiovascular disease treatment. In particular, it is reported that spirulina produces clionasterol, which protects vascular cells. Microalgae produce many antioxidant compounds, such as astaxanthin, beta-carotene, dimethylsulfonylpropionate, mycosporins, and some other carotenoids, which protect the organism from various diseases, aging, and oxidative stress. In recent years, with the introduction of probiotic supplements, microalgae have been investigated more intensively in the diet

in relation to their benefits for human health. When comparing the quality of proteins, microalgae biomass excels over vegetables, rice, and wheat, but is diminished in relation to milk and meat. Of a wide range of microalgae that produce bioactive compounds for commercial purposes, the most important are *Dunaliella*, *Haematococcus*, *Nannochloropsis*, *Chlorella*, and *Chlamydomonas*, and the cyanobacterium *Spirulina*. Several species of spirulina, such as *Spirulina platensis* and *Spirulina maxima*, are grown commercially, due to their protective or therapeutic role in the immune system, arthritis, anemia, cardiovascular disease, diabetes, and cancer. In addition, lactic acid bacteria in the gastrointestinal tract also increase, resulting in an improvement in the hormonal system.

Chlorella microalgae produce biomass that is rich in protein (51–58% dry weight), carotenoids, and various vitamins. *Chlorella* is also used to produce *Chlorella* growth factor, which improves the growth of lactic bacteria in the body and acts as a growth factor for intestinal bacteria. Many positive health effects from the use of *Chlorella* are recorded, such as increasing the level of hemoglobin, lowering blood glucose and cholesterol, and hepatoprotective action. *Chlorella* species produce metabolic products that can activate the immune system and stop *Candida albicans* and *Listeria monocytogenes* from growing. *Chlorella* extracts are also used in mammals to produce splenocytes and cytokines and to activate other immune responses.

Dunaliella microalgae (especially *D. salina*) are rich in lipids and proteins, glycerol, and beta-carotene, and have particular growth potential in brackish waters. Carotenoids produced by both *Dunaliella* and *Spirulina* species are more effective against cancer cells compared to beta-carotene.

In China, *Nostoc sphaeroides* has been used in traditional medicine to treat diarrhea, hepatitis, and hypertension. *Muriellopsis* sp. also produces a high level of carotenoids, such as lutein used to treat degenerative diseases. The cyanobacterium *Nostoc* has been a food source since 2000 in China and is considered healthy in China, Korea, and Japan due to its high protein content [21].

4.1. Commercially Available Products

The precursor to vitamin A, beta-carotene, is a useful natural food coloring and part of a balanced diet. It is also a necessary nutrient and possesses antioxidant qualities. It is lipophilic and non-polar, and soluble in carbon disulfide, benzene, and chloroform, but insoluble in water, acids, and alkalis. Every year, 8.5–30% of the beta-carotene needed for human consumption worldwide is produced commercially from the salt-tolerant microalgae *Dunaliella salina*. The market for beta-carotene reached USD 334 million globally in 2018. Australia, Israel, India, and China are experiencing an increase in beta-carotene production and commercial activity [2].

A natural source of pigments with antioxidant qualities and benefits for human health is the recombinant protein astaxanthin, a type of carotenoid. Specifically, it helps treat resistance to infections, coronary heart disease, and skin cancer. By 2020, the astaxanthin market is expected to grow to a value of USD 1.1 billion globally. When heated, astaxanthin dissolves in acetone, acetic acid, chloroform, pyridine, and DMSO. It is also polar and lipophilic. Astaxanthin can be accumulated by a variety of microalgae, including different strains of *Chlorella*. However, *Haematococcus pluvialis*, which is grown in indoor and outdoor PBRs or hybrid PBR systems for infection control, is the primary source used in its commercial production [2]. Astaxanthin, unlike other carotenoids, is not a precursor to vitamin A. Its synthesis for commercial purposes is undertaken using petrochemical raw materials, but this makes it prohibitive for human consumption, as defined by the FDA, and thus enhances the need for natural astaxanthin production. Microalgae that produce astaxanthin as a secondary metabolite are an increasingly important source of natural production, the most important being *Haematococcus pluvialis*. Its cells accumulate the highest amount of astaxanthin, which makes up 90% of all its carotenoids and can reach 7% on dry biomass (Rammuni et al. [29]). Beta-carotene and astaxanthin show strong demand growth in the market. The commercial value of carotenoids was anticipated to

increase at an annual growth rate of 3.5% from 2016 to 2021, reaching revenues of EUR 1.52 billion, according to forecasts made by Zion Market Research Global in 2016. By 2021, it was anticipated that astaxanthin and beta-carotene would account for 25% and 26% of the market, respectively. The microalgae *Haematococcus pluvialis* and *Dunaliella salina* are promising sources of carotenoids for industrial processing due to their ability to accumulate carotenoids, and because of the projected increase in market demand [29].

Fish oils contain eicosapentaenoic acid (EPA), an omega-3 fatty acid with positive effects on human health. The global market for omega-3 fats was valued at USD 34.7 billion in 2016. Fish oil is the primary source of EPA, but *Nannochloropsis* species offer a promising substitute because, depending on growing conditions, they may produce EPA at levels of 1.1–12% dry weight. Commercially, *Nannochloropsis* sp. are grown photo-autotrophically in tanks with natural light and CO₂ from power plants, but heterotrophic or mixotrophic cultures can be grown at the pilot or small stage [2].

It is also possible to add microalgae as a feed supplement. Because it has been shown to be advantageous to animal physiology, the species *Arthrospira*, in particular, is used as feed for aquarium fish, cattle, cats, dogs, horses, and shellfish. Although there are drawbacks, microalgae are also utilized as a source of protein in poultry feed, although not in higher concentrations or for extended periods of time. The primary side effects of feeding microalgae to poultry are changes in the skin and egg yolk colors. Over 50% of *Arthrospira* cultivation is grown for feed supplement production, and 30% of algae production worldwide is used to produce animal feed [21]. Because spirulina contains high percentages of protein (60–70% dry weight), vitamins, and minerals, it is also used in animal feed. The major producers in the world in 2010 were Chile, China, India, and the United States, with an annual production of 5000 tons. Alkaline and highly salinized conditions are used for commercial production; the biomass is subsequently gathered and processed for use in the food industry [2].

Extracts of *Nannochloropsis*, *Chlorella*, *Spirulina*, and other microalgae have all been utilized in skin and hair care products. To specifically guard against UV radiation damage and hyperpigmentation, dermal cosmetic products should contain carotenoids like lutein, beta-carotene, and astaxanthin. Likewise, cosmetic products can incorporate polysaccharides derived from different species of green microalgae to achieve antioxidant activity, gelling, or thickening effects [2].

Cosmetics companies have their own microalgae production system. Microalgae ingredients are usually applied as antioxidants, and as water-binding agents in cosmetic production. Protulines act against aging. *Dermochlorella* polysaccharide extract from *Chlorella vulgaris* enhances skin collagen production, resulting in tissue revival and reduced wrinkles, and is used by cosmetics companies. Other products developed containing *Nannochloropsis oculata* have excellent epidermal effects, and a product containing *Dunaliella salina* promotes cell growth and enhances the skin's energy metabolism [21].

4.2. Emerging Products

The metabolic potential of microalgae offers new opportunities to increase the range of products produced. Research is primarily directed towards the use of microalgae for the development of methods of expression and accumulation of recombinant proteins and biopolymer production.

Microalgae have been used to express recombinant proteins (RPs), including industrial enzymes, immunotoxins, antibodies, and vaccine subunits. Although most research has concentrated on *Chlamydomonas reinhardtii*, *Chlorella* and *Dunaliella* species are also thought to be safe organisms that can be utilized for RP production. A big single-chain antibody against the glycoprotein of the herpes simplex virus was the first antibody to be expressed. Recently, the chloroplast of *Chlamydomonas* has been shown to express single-chain mono- and bilateral immunotoxins, in addition to the complete human immunoglobulin G (IgG) antibody against anthrax. *Chlamydomonas reinhardtii* recombinant proteins have been extracted and purified using a variety of processing techniques [2].

The global market for bioplastics, or renewable, biodegradable plastics, is projected to grow to USD 10 billion by 2020. Microalgae biomass and polymers or additives are combined to form the final form of bioplastics, which are then formed into objects, sheets, and films. Microalgae bioplastics, also known as hybrid plastics, cellulose-based plastics, polylactic acid, or biopolyethylene, are utilized in the packaging, catering, gardening, medical, and automotive industries. The production of bioplastics from spirulina and chlorella is frequently employed because of their small cells and the proteins they produce, which enable conversion to bioplastics without the need for prior treatment. Bioplastics are produced from microalgae biomass and their proteins via a series of steps that include denaturation, digestion, fermentation, lamination, mixing, and compatibility. The production of bioplastics from microalgae is comparatively simple, and unlike other common bioplastic raw materials like soybeans, the use of microalgae has less of an impact on the food industry. The functional characteristics of microalgae proteins should be enhanced, and techniques for eliminating volatile compounds that cause odors should be developed, in order to attain commercial viability [2].

The global bioplastics production market is expected to reach 9.2 million tonnes in 2021, from 1.7 million tonnes in 2014. Polylactic acid (PLA) plays a leading role in biodegradable biopolymers. The properties of polylactate are comparable to those of conventional polymers produced from fossil fuels, such as polypropylene or polystyrene, so it is applied to a wide range of products (from packaging materials to medical applications). For the production of PLA, starchy materials such as corn and potatoes are used as raw materials, but they are foods, and therefore incompatibility is created in nutritional needs [30]. In addition, starch combines the advantages of low cost, abundance, and biodegradation, making thermoplastic starch (TPS) a dynamic alternative to fossil fuel plastics. Microalgae, with their increased ability to adapt to changing environmental conditions, can accumulate biopolymers, such as polysaccharides and, especially, starch, when cultivated under stress. In particular, the lack of essential nutrients, such as nitrogen and phosphorus, as well as the macro-element sulfur, is a factor that induces the storage of biopolymers in microalgae cells during their growth. The direction of research on microalgae that produce increased amounts of starch is of strategic importance, as the produced starch, in addition to its plasticization, can also be a substrate for fermentation for the production of bioethanol [31]. In this study, the possibility of increased starch production among ten strains of microalgae was examined. The initial selection for eight of them was based on their rapid and dynamic growth, while the other two were of the species *Chlamydomonas*, which have been extensively investigated for their starch-producing properties. Crop growth began with incubation in agar plates followed by inoculation of cultures in conical flasks of 125 mL of artificial substrate for crop growth. One of the two strains of *Chlamydomonas reinhardtii* provided satisfactory starch production, which reached 49% by weight of biomass, whose maximum production was 5.07 g/L. These performances were recorded on the bottles after 20 days and under conditions of sulfur deficiency. The selected crop of *Chlamydomonas reinhardtii* was transferred to a small-scale pilot-stage photobioreactor of 30 L, where the same numerical results were not achieved, as there was contamination by a strain of *Chlorella*. However, given that only the strain of *Chlamydomonas reinhardtii* accumulated starch under the sulfur deficiency conditions applied to the photobioreactor, the authors estimate that they reached the same percentages (54%) as those of the bottles. The biomass produced by the 30 L photobioreactor was led to lamination using 30% glycerol (*w/w*) at 120 °C, where through examination of the structural characteristics, the homogeneous lamination and consequently the possibility of producing bioplastics by microalgae was confirmed [31].

5. Potential for Concurrent Production of Biofuels

Microalgae, through their rapid growth, which goes hand in hand with their high CO₂ sequestration capacity and minimum land requirements, constitute a dynamic energy resource with a positive environmental impact and that does not compete with the food sector. With their capacity to provide biomass with increased amounts of carbohydrates,

proteins, and lipids, they can lead to the production of a range of biofuels [32]. The biofuels produced from microalgae biomass are third-generation biofuels and are an evolution of the first generation, which included the production of biofuels from edible crops, and the second generation, produced from lignocellulosic biomass originating from agricultural crop waste and, partly, the organic fraction of municipal solid waste. With the evolution of generations, the advantages and disadvantages mainly related to raw materials are changing.

Various microalgae species, such as *Chlorella vulgaris*, *Nannochloropsis* sp., and *Chlamydomonas reinhardtii*, have been extensively studied for the production of biofuels as they have recorded high growth rates and lipid contents compared to other microalgae species. *Chlorella*, *Chlamydomonas*, *Dunaliella*, *Scenedesmus*, and *Tetraselmis* have been found to have an increased content of polysaccharides (>40% on a dry basis), and their biomass can be a fermentation medium for bioethanol production. The necessary elements for the growth of microalgae, such as carbon, nutrients, and metals, are found in wastewater in increased quantities, thus ensuring cheap biomass for biofuel production with simultaneous bioremediation of waste. Of the biofuels produced from microalgae biomass (bioethanol, biobutanol, biodiesel, biohydrogen, and biomethane), biodiesel and bioethanol are those studied on larger production scales [33].

The raw materials most used worldwide for biodiesel production are rapeseed oil (59%), soybean oil (25%), palm oil (10%), sunflower oil (5%), and other sources (1%) such as coconuts, peanuts, corn, cooking oil, animal fats, and algae. Some microalgae species grown under optimized growing conditions have the potential to produce 47,000–141,000 L of lipids/ha/year, which represents an oil yield of more than 200 times that of the most productive plants. The predominant problem, however, for the industrial application of microalgae biodiesel production is that more energy is required to increase the energy requirement to increase biomass concentration than the energy produced by microalgae biomass [34]. Several microalgae species contain remarkable lipid amounts. Common microalgae, including *Chlorella*, *Dunaliella*, *Isochrysis*, *Nannochloris*, *Neochloris*, *Phaeodactylum*, *Porphyridium*, and *Schizochytrium*, have lipid contents ranging from 20 to 50% dry weight. These values can be increased by appropriate handling of growth factors, optimization of the cultivation and harvesting process, and genetic modifications of the selected strains. Nitrogen deficiency and increased salinity can lead to the induction of fatty acid growth in the majority of microalgae. A fatty acid consists mainly of glycerides, which are esters of glycerol (which has three -OH) with fatty acids, and are therefore called triglycerides (TAG). The fatty acid composition of most microalgae is dominated by the fatty acids C14:0, C16:0, C18:1, C18:2, and C18:3, but the relative composition varies from species to species. Lipids can be converted to fatty acid methyl esters (FAMES) through transesterification for biodiesel production. Transesterification is the most common technology for the production of biodiesel from biomass, where triglycerides react with short-chain alcohol (e.g., methanol or ethanol) in the presence of alkaline or acid catalysts. The main byproduct of the process is glycerol, which is used in various industrial applications. In addition, residual biomass after oil removal can be used as animal feed [35].

Microalgae biomass can be effectively used to produce biogas, methane, hydrogen, and bio-hythane (5–25%). The main disadvantage of anaerobic fermentation (AD) leading to biogas production is the difficulty of solution by enzymatic hydrolysis of the microalgae cell wall [35]. AD converts almost all organic components of biomass into biogas through complex microbial action, without drying or extraction processes. By applying hydrothermal pretreatment to biomass, an improvement in the efficiency of biofuel gas production has been found, not only on a laboratory scale but also on an industrial scale [32]. Despite this advantage, the use of microalgae biomass as a feedstock for AD faces a number of technological challenges, which, in addition to the resistance of the cell structure, include the presence of toxic substances in anaerobic bacteria. Pre-treatment of microalgae biomass (residual or total) prior to AD seeks to address these problems. The main methods applied are pre-treatment methods, such as mechanical methods (size reduction, ultrasonic, or

microwave), thermal hydrolysis, chemical methods (oxidation, addition of alkalis, acids), biological methods (enzymes), and hybrid methods where a combination of methods is performed. From the comparison of the above pre-treatments, thermal hydrolysis appears to have the highest Energy Return Over Investment (EROI), with the EROI ratio reaching 6.8. In addition to pre-treatment, recycling of AD outputs as inputs to AD has been proposed in order to make the system economically and environmentally sustainable [35].

Bioethanol production is increasing, from 17,250 mL in 2000 to 160,000 L in 2020. Microalgae are an alternative resource for bioethanol production due to their starch and cellulose contents, which can be broken down into fermentable sugars such as glucose. In addition, they have the advantage compared to lignocellular matter that no lignin is contained in their cell wall, thus facilitating its breakdown. Through the fermentation of sugars, bioethanol is produced. For the breakdown of carbohydrates into fermentable forms of simple sugars, a saccharification stage is applied before fermentation. The production of bioethanol by microalgae is greatly influenced by the pre-treatment and type of fermentation process. The main methods used for the pre-treatment of microalgae biomass in order to disrupt the cell wall to release more carbohydrates and convert them to monosaccharides are chemical methods, mainly acid hydrolysis and alkaline hydrolysis. For acid hydrolysis, sulfuric and hydrochloric acid have been applied, while for alkaline hydrolysis, caustic soda, soda ash, and ammonia solution have been applied. In the case of acid hydrolysis, inhibitors are formed which have a negative effect on the fermentation process, and therefore neutralization must be carried out which further increases the production cost of bioethanol. Enzymatic hydrolysis is more efficient and has lower energy needs compared to chemical hydrolysis, although the cost of selected enzymes is high, thus limiting its commercial application. The choice of pre-treatment technology is closely correlated with the characteristics and efficacy of enzymes. Mechanical methods such as crushing, grinding, or milling, as well as the use of ultrasound, increases the accessibility of the substrate to enzymes, but have high cost and energy requirements compared to chemical methods. In addition, a combination of methods, such as the use of ultrasound with enzymatic hydrolysis, can be used with satisfactory results [36]. The microalgae strains that have been most scientifically investigated for increased carbohydrate production are *Isochrysis galbana*, *Porphyridium cruentum*, *Nannochloropsis oculata*, and species of *Spirogyra* and *Chlorella* [35].

Biobutanol is considered a more efficient biofuel than biomethanol or bioethanol due to its higher energy density, and is compatible with gasoline due to its similar molecular structure. In addition, it has the advantage of requiring almost half the heat of evaporation compared to ethanol, and this is an advantage when starting the engine at sub-zero temperatures. Due to these properties, it can be used, in internal combustion engines, either as a mixture with diesel (good solubility) or alone as fuel. It is easier to store and transport due to the low vapor pressure and is also easier to manage. Biobutanol, apart from being a fuel, also finds many applications in the pharmaceutical and food industries. Its production is undertaken via anaerobic fermentation of acetone, butanol, and ethanol (ABE), with the bacterium *C. acetobutylicum* and substrate carbohydrates of microalgae biomass. In the first step, butyric and acetic acid are formed, the pH is reduced, and then acetone, butanol, and ethanol are produced in a ratio of 3:6:1. Because this bacterium is a saccharolytic, no pre-treatment step is required for fermentation as is done for ethanol production, and this is a significant economic advantage. Microalgae biomasses such as *Tetraselmis subcordiformis*, *Chlorella vulgaris*, *Chlorella reinhardtii*, and *Scenedesmus obliquus*, can be suitable substrates for investigating biobutanol production as they are characterized by high starch content and convertible sugars [37].

Compared to other fuels, hydrogen (H₂) has the highest energy content (142 kJ/kg), is safer to handle than natural gas, and can be transported via conventional means for use in industry or cities. Furthermore, because it is the only fuel devoid of carbon, all that is produced when it oxidizes is water. Coal, oil, natural gas, and naphtha, along with very little water and biomass, are used as fossil fuels to produce hydrogen. These procedures are

expensive and polluting. As a result, there is a growing interest in producing biohydrogen (BioH₂). Microbial electrolysis converts wastewater into BioH₂. Anaerobic microorganisms can also produce BioH₂ through fermentation processes like dark fermentation, which involves converting sugars into H₂, CO₂, and organic acids. High rates of hydrogen production, the capacity to produce hydrogen from biomass or biowaste, and the accomplishment of efficient process planning and control are significant benefits of this process. Because they can absorb solar energy and CO₂, and transform them into storage chemicals like starch, a fermentable substrate for the production of BioH₂, microalgae biomasses make excellent raw materials for dark fermentation.

BioH₂ can be created under more controlled fermentation conditions using particular strains of microorganisms that produce higher amounts of hydrogen, or from mixed cultures of microorganisms where raw materials are waste. These microorganisms include thermophilic bacteria like *Thermotoga neapolitana* and mesophilic bacteria like several species of *Clostridium* and *Enterobacter*. In fermentations using microalgae biomass feedstocks like *Anabaena* sp., and microalgae strains like *Chlamydomonas reinhardtii*, *Nannochloropsis* sp., *Chlorella vulgaris*, and *Scenedesmus obliquus*, which grew on an artificial substrate and municipal waste, these anaerobic bacteria achieved high hydrogen productivity [38]. The production of hydrogen by microalgae depends on the activity of the enzyme hydrogenase, which is particularly sensitive to the presence of oxygen. To achieve anaerobic conditions for hydrogen production, the oxygen produced must be removed. Therefore, strictly anaerobic conditions are necessary for the efficient production of hydrogen from microalgae. This phenomenon present in natural bacteria–microalgae symbioses can also be achieved in artificial bacteria–microalgae cultures, and lead to higher hydrogen yields. This was demonstrated by the symbiosis of the bacterial strains *Brevundimonas* sp., *Rhodococcus* sp., and *Leifsonia* sp. with the cultivation of *Chlamydomonas*, where hydrogen production was enhanced. In addition, hydrogen production from *Chlamydomonas* was significantly enhanced when it was placed in artificial symbiosis with a modified strain of hydrogenase *Escherichia coli*. In conclusion, microalgae can capture light energy and produce hydrogen, with the synergy of bacteria that remove oxygen produced during photosynthetic activity, without the use of exogenous organic substances or other nutrients [39].

Moreover, biochar is the solid, carbon-rich material produced by pyrolysis of any form of organic biomass (forestry, agricultural crop residues, paper sludge, and poultry farm waste). The pyrolytic process involves heating between 350 °C and 700 °C in the absence of oxygen. The creation of biochar from microalgae biomass is a carbon capture tool [40]. Biochar is considered a “green” product with the potential to be used in a sustainable strategy for the development of negative carbon emission technology. The conversion of biomass to biochar plays an important role in carbon capture and is an alternative approach to the energy utilization of biomass by capturing and storing CO₂, resulting in the production of biofuels with NET (negative emission technology) [41]. Biofuels remove carbon from the atmosphere as the raw materials for their production photosynthesize to grow, as well as through biochar production, as an amount of carbon returns to the soil when it is used as a soil improver. The ratio of 75:25 for biofuel production:biochar, is considered the most ideal, as by increasing the ratio to 90:10, the production of biofuels is maximized but the negative carbon footprint is reduced due to the reduction of biochar. Similarly, a 60:40 ratio would create the reverse effect. The strategic regulation of biofuel production in relation to biochar production is critical to our carbon footprint. Therefore, in optimizing the production of biofuels from microalgae, it would be environmentally important to add the production of biochar, as through the pyrolysis of microalgae biomass, energy production would be achieved while at the same time the biochar produced would capture a large percentage of carbon [40]. The characteristics of biochar, such as pH, porous structure, specific surface, bound carbon, ash, and ion exchange capacity, are affected by pyrolysis conditions, with the temperature and type of biomass being the most decisive. Biochar produced from microalgae is poorer in carbon, surface area, and ion exchange capacity, but with increased pH and minerals (P, Ca, K, and Mg) compared to biomass from lignite

cellulosic materials [40]. Biochar can be used for heat and energy production, as a carbon sequestration agent, as an adsorbent in various industrial applications, and as a fertilizer. As a fertilizer, biochar has the potential to gradually release key nutrients (N and P), which enhance plant nutrition efficiency and consequently increase crop yields. In addition, biochar can improve the physicochemical characteristics of the soil, such as increasing residual nitrogen and carbon in the soil, improving the soil's pH and electrical conductivity. In biochar production, evaluation of its characteristics is important to determine its potential applications. Biochar must comply with certain specifications to be used as fertilizer, such as a C content of more than 50% on a dry basis, with a range of N and P from 1% to 45% (the usual values are 15% for P and 1% for N), pH up to 10, and a specific surface area greater than 150 m²/g. [42]. In the study of Yu et al. [41], pyrolysis was applied in a stainless steel cylindrical reactor, using biomass produced by *Chlorella vulgaris* FSP-E cultivated on modified basal medium substrate, in a photobioreactor with a continuous flow of 2.5% CO₂ and a temperature of 26 ± 1 °C. Pyrolysis was carried out in a stainless steel cylindrical fixed-bed reactor at a temperature of 500 °C with a heating speed of 10° C/min, with a continuous flow of nitrogen of 100 mL/min. The authors concluded that biochar produced from *Chlorella vulgaris* FSP-E in these specific conditions is an excellent soil improver. In conclusion, the results of the study show that the cultivation of *C. vulgaris* FSP-E microalgae and biochar production, through pyrolysis, can be used as a potential clean technology for carbon capture in the context of a microalgae biorefinery for a sustainable environment [41].

6. Specific Applications of Wastewater Bioprocessing with Simultaneous Biofuel Production

6.1. Municipal Wastewater and Bioethanol/Biodiesel Production

In the context of investigating the production of bioethanol from microalgae in municipal effluent substrate, the study of Onay [43] evaluated *Hindakia tetrachotoma* ME03 at various wastewater concentrations. The waste, after sterilization, was mixed with Bold's Basal Medium (BBM) in concentrations of 0%, 25%, 50%, 75%, and 100%. The maximum biomass productivity (0.097 g/L/d) was observed in the 25% sample, but the 50% sample had the highest specific growth rate (μ) = 4.89/d, compared to the value of the 25% sample of 4.74/d, with slightly lower biomass productivity (0.090 g/L/d). In addition, the 50% sample had the highest carbohydrate content (23.6 ± 0.4%). With these criteria, the 50% waste concentration sample was selected to be tested for bioethanol production through fermentation. The maximum bioethanol concentration (11.2 ± 0.3 g/L) was observed at 36 h of incubation with *Saccharomyces cerevisiae*, with a percentage yield of 94 ± 2.2%. The conclusions of the study showed that the selected strain *H. tetrachotoma* ME03 can be considered a type of mycoalgae for bioethanol production in municipal wastewater substrate, giving satisfactory results, after solving bacterial and fungal contamination problems through specialized technologies [43]. Satisfactory yields in biodiesel production and simultaneous treatment of municipal waste are recorded in the study of Tripathi et al. [44]. The effluent samples examined came from inputs to wastewater treatment facilities near New Delhi, India. They were diluted to concentrations of 25, 50, and 100%, after being sterilized with synthetic substrate BG-11. The strain used was *Scenedesmus* sp. ISTGA1, isolated from the rock surface of a marble mining area adjacent to New Delhi. The crops lasted 14 days and the results showed satisfactory removal of heavy metals and a plethora of organic substances. The strain performed better on all three wastewater samples than on the reference sample artificial substrate (BG-11). Indicatively, concentration reductions of 91.50, 92.09, and 82.30% were recorded for Zn, Fe, and Al, and complete removal of Cd, Ni, Pb, and Co. At the same time, there were large decreases in BOD and COD values, of 86.74% and 88.82%, respectively, while nitrates and phosphates were completely removed. In relation to lipid production, the maximum production was achieved by the 100% waste sample. According to the above, the strain *Scenedesmus* sp. ISTGA1 can be used for the development of municipal effluent bioremediation technology with simultaneous sustainable biodiesel production [44]. The study by Amit and Ghosh [45] evaluated the potential

of microalgae of marine origin of *Tetraselmis indica*, in relation to three other freshwater plants, for the treatment of municipal waste for the production of fatty acids for conversion to biodiesel with simultaneous bioremediation. Samples were taken from primary and secondary treatments of local municipal wastewater treatment facilities at a university site in India and the crops lasted 14 days. Bioremediation in all samples was satisfactory, as both organic carbon (71.16–85.70%) and the nutrients nitrogen (63.6–78.24%) and phosphorus (60.90–65.97%) were removed. The measurements showed that *Tetraselmis indica* had the most satisfactory yields compared to the others, and the highest biomass yield amounted to 0.6533 g/L, while lipid productivity amounted to 25.44 mg/L/d, in the secondary sample. The profile of fatty acids produced from the biomass met the specifications for biodiesel production and use in internal combustion engines, as it included myristic (1.2%), pentadecyclic (0.28%), palmitic (10.32%), oleic (34.59%), linoleic (12.38%), and eicosanoic acids (14.88%) [45].

6.2. Dairy Wastewater and Biodiesel Production

Lipid production with simultaneous bioremediation of effluents and CO₂ capture, through appropriate microalgae is also achieved in highly burdened waste such as that of dairy farms. In the study of Mousavi et al. in 2018, a new strain was isolated from a local cattle dairy farm (Babolsar, Iran), *Coelastrum* sp. The waste used for the cultivation of the microalgae was the output from the solid material separator of the waste treatment plant and was burdened with COD: 34.949 (mg/L), TKN (Kjeldahl nitrogen): 2794 (mg/L), and TP (total phosphorus): 316.67 (mg/L). *Coelastrum* sp., after a latency that lasted two days, adapted to waste conditions and entered an exponential growth phase. The experiments were conducted in four COD concentrations (600, 750, 900, and 1050 mg/L), where it was found that the optimal concentration for biomass production and CO₂ capture was in the COD sample of 750 mg/L. Biomass productivity reached 0.266 g/L/d and CO₂ capture reached 53.12 mg/L/d. In addition, the mean elimination efficacy in optimal conditions was COD: 53.45%, TKN: 91.18%, and TP: 100%. Simultaneously with the above, the study investigated the effect of light, as light intensity is one of the main factors affecting the growth and accumulation of lipids in microalgae biomass. The light intensity in optimal biomass growth conditions was 6900 Lux, while for lipid production, measurements were made under different lighting intensities and the maximum lipid productivity reached 11.08 mg/L/d under 2300 Lux lighting. The explanation for the fact that optimal lipid production was at a lower light intensity than biomass production is due to the dependence of lipid synthesis on chlorophyll, through ATP and NADPH photosynthetic action. The production of chlorophyll by chloroplasts decreases when light is unlimited, but at high light intensities chlorophyll can be degraded, resulting in a decrease in production. In conclusion, the results of the study, according to the authors, demonstrated that *Coelastrum* sp. is a suitable strain for wastewater bioremediation, binding CO₂, and fatty acid production [46].

6.3. Livestock Waste and Bioethanol Production

In addition to microalgae that accumulate lipids in their biomass when grown in livestock waste, there are also species that accumulate increased amounts of carbohydrates and may be utilized as a fermentation feedstock for bioethanol production. Pig farm waste (manure) is one of the most heavily burdened wastes worldwide due to its high concentration in organic load and greenhouse gas emissions. In Mexico, about 50% of methane and nitrogen oxide emissions come from the decomposition processes of this waste. Due to the very high concentrations of total nitrogen (9000 to 18,000 mg/L) and total phosphorus (6200 to 39,000 mg/L), anaerobic digestion is the best available technology for the management of wastewater from pig waste. From this process, biogas and digestion residue is produced, which is a liquid rich in nutrients (nitrogen and phosphorus) and organic acids, and can therefore be used for microalgae cultures [47]. The green microalgae *Chlorococcum* sp. was isolated from waste treatment plants in Mexico and cultivated

in anaerobic digestion residue of pig farm waste via two approaches. Firstly, growing conditions were controlled for lighting and temperature, while secondly, there was no intervention to control conditions. In the 27-day controlled culture, lighting was maintained consistently at $46 \pm 3 \mu\text{mol photons/m}^2/\text{s}$ and temperature at 30°C . In the uncontrolled culture, lighting ranged from $94 \mu\text{mol photons/m}^2/\text{s}$ to $357 \mu\text{mol photons/m}^2/\text{s}$ with an average of 24 days of cultivation, $98 \mu\text{mol photons/m}^2/\text{s}$. In addition, the temperature inside the greenhouse ranged from 25°C to 39°C , with an average of 30°C . The results showed that under controlled conditions, the highest biomass production was found in the 8% digestate sample and amounted to 0.85 g/L , while in the uncontrolled conditions it reached 0.64 g/L in the 5.6% digestate sample, which is very close to the 0.64 g/L reached in the reference sample (modified Bold's Basal Medium). In relation to lipid formation, controlled cultures ranged from 5.4 to 9.8% on a dry basis, while non-controlled cultures ranged from 9.6 to 12.9% with a content of C18:3n6 (gamma-linolenic acid), which exhibited very high levels (26–35%). At the same time, while the carbohydrate content from the inoculum stage was 198 mg/g , it reached $454.1 \pm 2 \text{ mg/g}$ on a dry basis, a value higher than the reference sample ($425.6 \pm 10 \text{ mg/g}$), with the increase occurring when ammoniacal nitrogen declined. The FAME profile in the two cases was different and did not meet the requirements of EN 14214 for biodiesel production. On the contrary, it was found that the strain *Chlorococcum* sp., under conditions of nitrogen deficiency from the 12 days of cultivation, accumulated carbohydrates and not lipids. The same phenomenon has been observed in cultures of *Chlorella vulgaris*. According to the above, the authors consider that the cultivation of *Chlorococcum* sp. under uncontrolled conditions, in a properly diluted anaerobic digestion residue of pig farm waste, is a promising system for bioethanol production within a microalgae biorefinery. The lipids produced can potentially be high-added-value products, since gamma-linolenic, palmitic, and linoleic acids, accounting for about 40–60% of the total lipid content of the biomass of *Chlorococcum* sp., are of great value in aquaculture [47].

6.4. Food Processing Plant Waste and DHA Production

In addition to biofuels, microalgae also produce polyunsaturated fatty acids (PUFAs) that are of significant nutritional value. One of these is docosahexaenoic acid (DHA), whose main production is based on fish oils (salmon, tuna, sardines). The main disadvantages of fish oil production are sustainability issues due to seasonal and climatic variations, overfishing, and depletion of fish populations. In addition, however, heavy metals (mercury) and other dangerous pollutants (PCBs and dioxins) have been tracked at elevated levels in some fish species. In addition, in relation to the production of fatty acids, there is a significant difference between those produced from fish oil and those produced from microalgae. In the first instance, extra separation and purification procedures are needed during DHA production in order to comply with industry standards. However, due to their lower trophic levels than fish and their status as primary producers, microalgae have far simpler fatty acid profiles, which facilitates separation [48]. In their 2018 study, Park et al. examined the economic feasibility of producing DHA by cultivating the strain *Aurantiochytrium* sp. KRS101 on an orange peel waste (OPW) nutrient substrate that was acquired from a Korean orange juice factory. Orange is the most-used fruit in the fruit juice industry; 51.1 million tons of orange juice are produced annually worldwide. The biomass of the fruit, which is made up of solids like pulps and peels, makes up about half of its byproducts when juice is produced. As a result, millions of tons of orange peels could be made available for use as a microalgae growth substrate, although substrates such as anaerobic digestion effluent (ADE) are found to be effective, especially in carbohydrate level augmentation [49]. Although there is more potential for other uses, the primary use is as inexpensive feed. Orange peel extract (OPE) was optimized in the study using NaNO_3 (1.2 g/L) as a nitrogen source, which produced higher yields of DHA (0.63 g/L) than urea and NH_4Cl . The yield of $0.63 \pm 0.02 \text{ g/L}$ is 2.5 times that of the modified standard synthetic substrate and was

recorded at 72 h of cultivation. At the same time, there was a high performance in FAME (1.97 ± 0.07 g/L).

To investigate economic viability, the authors calculated production costs comparatively between the typical synthetic substrate and the OPE with the addition of sodium nitrate. The cost of the synthetic substrate is 25.75 USD/1000 L, while that of the OPE is 3.52 USD/1000 L. Therefore, using *Aurantiochytrium* sp. KRS101 on OPE substrate and 1.2 g/L NaNO_3 to produce 1 kg of DHA is 18 times less expensive than using modified basal medium with 20 g/L glucose and 2 g/L yeast extract. Given that OPE produced 0.63 g/L of DHA and the modified synthetic medium produced 0.25 g/L, the raw material costs per kg of DHA for OPE and the synthetic medium, respectively, were USD 5.6 and USD 103. According to the authors, the findings of the study support the effective exploitation of waste from food processing plants for microalgae crops that produce value-added products, such as DHA [48].

6.5. Waste from the Munitions and Explosives Industry

In the study of Abraham et al. [50], it was proposed to use microalgae cultures to recover nutrients from untreated waste from the munitions and explosives industry. This method permits the production of renewable energy sources, such as biodiesel or biogas, while also lowering the concentration of nutrients. The production of explosives, or active compounds, for the armaments industry, such as 3-nitro-1,2,4-triazol-5-one (NTO) and 1,3,5-tritroperohydro-1,3,5-triazine (RDX), generates large amounts of waste that contain organic pollutants that are rich in nutrients, primarily nitrogen. This wastewater requires physicochemical and/or biological treatment before being released into water bodies. Five species of laboratory microalgae (two freshwater and three seawater), and *Scenedesmus obliquus* (ATCC[®] 11477), were selected. Ten munitions production plant sites were sampled for untreated waste, with varying levels of organic and inorganic carbon and nitrogen content. Only three of the waste streams needed chemical treatment because of their high concentrations of growth inhibitors, according to the results, while the majority of the streams could be used as microalgae culture nutrients without pre-treatment. In particular, *Scenedesmus obliquus* (ATCC[®] 11477) had similar growth in a mixture of untreated wastes with added CO_2 (atmospheric or combustible), relative to the reference sample (synthetic substrate). In conclusion, the study demonstrates that by applying microalgae cultures, bioremediation of the burdened waste of this industry can be achieved, while reducing its energy footprint in a sustainable and efficient way [50].

6.6. Anaerobic Treatment Residues of Organic Waste

As consumption of meat and animal products is expected to double by 2050 due to global population growth, the need to increase feed production will intensify, and this will lead to impacts on the food chain and greenhouse gas emissions. The management of organic food and feed production waste is a sector with ever-increasing demands. Anaerobic treatment is the dominant way of dealing with this issue, as it produces renewable energy and fertilizers while reducing emissions from waste storage. In particular, anaerobic treatment produces biomethane and a nutrient-rich residue, which is applied to agricultural crops, ensuring waste stabilization, reduction in greenhouse gas emissions, reduction in odor, and provision of nutrients necessary for crop growth. The residue is rich in macroscopic nutrients (N, P, K, S, Mg, Ca, Fe, and Na) and contains several trace elements (Co, Fe, Se, Mo, and Ni), either as a result of the original raw material used or due to appropriate additions for improved anaerobic treatment efficiency. It can be used in soils, untreated or after treatment, but it can also be subjected to separation of liquid and solid fractions. Filtration and dehumidification are among the most common treatment methods used depending on the needs of residue application and depending on the type of waste generated. However, the nitrogen content of the residue is quite enhanced and the form in which it is found is very easily absorbed by plants. This property creates a problem in areas rich in nitrogen, and therefore limits and hinders its application. Microalgae crops are one of the

options for making more efficient use of anaerobic digestion residue to avoid burdening soils with nitrogen loads and producing protein feed with low greenhouse gas emissions. Microalgae need nutrients to grow, and can therefore derive them from the residue of anaerobic treatment. The biomass produced is high in protein and can be used as a source of feed or fish feed. This approach creates a circular economy solution for organic waste that will reduce the impact of agricultural crops by drastically reducing nutrient pollution, greenhouse gas emissions, and the requirement for land use change in order to boost feed production, thereby ensuring the demanded food security [51] (Figure 2).

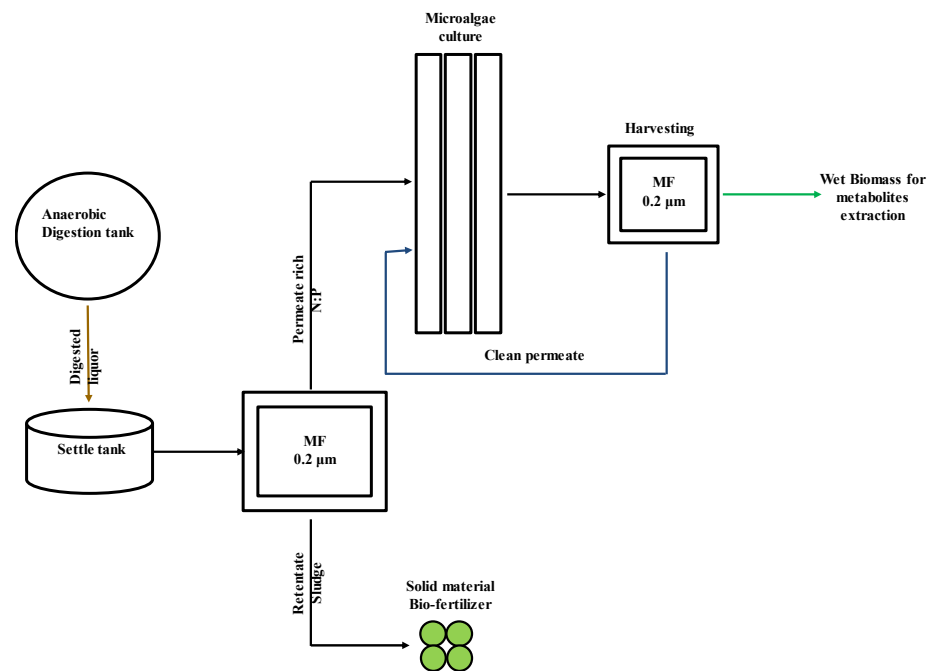


Figure 2. Simplified flow chart of anaerobic digestion residue treatment with microalgal cultures (modified from Stiles et al. [51]).

6.7. Recapitulation Table for Microalgae-Based Effluent Treatment

As can be seen from Table 4, the removal of organic materials, nutrients, minerals, and APIs using microalgae cultures is sufficiently satisfactory, while in all cases there is production of products from microalgae biomass, which depends to a large extent on the selected strain, the type of waste, and the pre-treatment they underwent to form the culture medium.

Table 4. Recapitulation table for wastewater treatment with microalgae.

Type of Waste	Types of Microalgae	Reduction in Waste Burden	Biomass Production/Productivity	Product Production	Source
Municipal waste	Hindakia tetrachotoma ME03		0.097 g/L/d (25% waste)	Bioethanol (11.2 ± 0.3 g/L (50% waste))	[43]
Municipal waste (Stream of inflow of treatment plants)	Scenedesmus sp. ISTGA1	Zn: 91.50%, Fe: 92.09%, Al: 82.30%. Cd, Ni, Pb and Co: 100%. BOD: 86.74% COD: 88.82%	1.81 g/L (100% waste)	Lipids (452 mg/L)	[44]
Municipal waste (Primary and secondary processing)	Tetraselmis indica	TOC: 71.16–85.70% N: 63.6–78.24% P: 60.90–65.97%	0.6533 g/L (secondary)	Lipids (25.44 mg/L/d)	[45]

Table 4. Cont.

Type of Waste	Types of Microalgae	Reduction in Waste Burden	Biomass Production/Productivity	Product Production	Source
Livestock waste (Cattle—output from solid material separator)	<i>Coelastrum</i> sp.	COD: 53.45% TNK: 91.18% TP: 100%	0.266 g/L/d (COD: 750 mg/L)	Lipids (11.08 mg/L/d)	[46]
Livestock waste (Pig farm)	<i>Chlorococcum</i> sp.		0.85 g/L (8% anaerobic digestion residue) on 454 mg/g carbohydrates on a dry basis	Lipids 9.8% dry (for use in animal feed)	[47]
Food processing plant waste (orange juice) (Study for nitrogen sources)	<i>Aurantiochytrium</i> sp. KRS101		4.2 g/L	DHA 0.63 g/L Lipids 1.97 g/L	[48]
Municipal waste *Study for removal of 19 APIs	<i>Chlorella</i> species <i>Coelastrum</i> sp. <i>Coelastrum astroideum</i> <i>Desmodesmus</i> sp. <i>Scenedesmus</i> sp.	Water soluble 40% Lipophilic 88%	2.6 and 5.4 g/L on a dry basis (<i>Chlorella vulgaris</i> 13-1 and <i>Chlorella saccharophila</i> RNY, respectively) with the best elimination yields for the 19 APIs.	Lipids 13–30% on a dry basis	[26]
Urban and rural co-management (Large-scale project in progress)	Mixed crops	Phosphate: 2 mg/L, Ammonia: 10 mg/L Nitrite: 15 mg/L Nitrate: 15 mg/L	2.2 kg VSS/d		[23]
Slaughterhouse waste (HRAP 75 L)	Mixed crops	N: 80.2% P: 70.8%	12.7 g VSS/m ² /d		[22]
Municipal waste (HRAP 530 L)	<i>Scenedesmus obliquus</i>	N: 65.1 ± 2.9% P: 58.8 ± 1.2%	245.5 ± 32.4 mgSS/L		[22]
Dairy waste (Tanks 100 L)	<i>Chlorella zofingiensis</i>	N: 97.5% P: 51.7%	0.175 g/L		[22]
2-stage municipal waste (Tanks 700–800–850 L)	<i>Scenedesmus</i> sp.	N: 79 ± 1% P: 57 ± 12%	17 ± 1 g/m ² /d		[22]
1-stage municipal waste (Tank 8 m ³)	<i>Mucidosphaerium pulchellum</i>	N: 74 ± 2% P: 79 ± 22%	17.4 ± 3 g/m ² /d		[22]
1-stage municipal waste (Tank 60 L)	Mixed crops	N: 92.7 ± 5.8% P: 82.9 ± 8.6%	1.7 ± 0.6 g/L/d		[22]
Aquaculture Waste (Tank 8 L)	Mixed crops	N: 83 ± 10 P: 94 ± 6%	5 g/m ² /d		[22]
Municipal waste (Settling tank) PBR 48 L	<i>Chlorella vulgaris</i>	N: 78.4 ± 8.2 P: 88.5 ± 4.5%	2.75 ± 0.2 g/L/d		[22]
Starch industry waste (Anaerobic effluents) PBR 890 L	<i>Chlorella pyrenoidosa</i>	N: 83.1% P: 68%	0.37 g/L/d		[22]
Municipal waste (Fabric membrane filtrate) PBR 60 L	<i>C. vulgaris</i>	N: 78% P: 78%	0.6 g/L		[22]
Municipal waste (Fabric membrane filtrate) PBR 60 L	<i>Scenedesmus obliquus</i>	N: 67% P: 67%	0.5 g/L		[22]
Pigsty waste (Anaerobic effluents) PBR 48 L	<i>S. obliquus</i>	N: 746 ± 6.9% P: 81.7 ± 7.3%	311.3 ± 12.4 mg/L/d		[22]
Aquaculture Waste PBR 4–40–400 L	Microalgal-bacterial floc	N: 4 L: 58 ± 11 40 L: 41 ± 8 400 L: 29 ± 25 P: 4 L: 89 ± 5 40 L: 65 ± 15 400 L: 59 ± 13	4 L: 109 ± 30 40 L: 45 ± 6 400 L: 11 ± 12 mg VSS/L/d		[22]

7. The Concept of Microalgae Biorefineries

A biorefinery is a sustainable biomass processing facility that combines low-impact chemical and biological technologies to produce a variety of high-value products and energy. Biorefineries encourage the production of goods and energy from renewable sources, which helps to achieve the Sustainable Development Goals. A biorefinery can be a concept, a manufacturing process, a factory, or even a collection of facilities, according to the definition. The fundamentals involve converting biomass into diverse product streams and integrating different technologies and processes in the most environmentally friendly manner possible. One of the most promising feedstocks for the production of biofuels, such as biodiesel, biomethane, biohydrogen, and bioethanol, is microalgae biomass. Additionally, it offers substitutes for the production of goods used in the food, feed, cosmetic, and health sectors, as well as for the synthesis of soil nutrients and bioplastics.

In contrast to the energy crop and oil industries, the microalgae biorefinery is a relatively new technology. However, with the aid of suitable genetic modifications, the addition of a novel technology combination, the application of nanotechnologies, the utilization of several valuable microalgae bioproducts, and the adaptation and improved comprehension of their cell biology, the commercialization and efficiency of the production of value-added products from microalgae biorefineries become feasible [21]. More specifically, microalgae biomass can be the raw material for the production of many bio-based products, such as fuels, chemicals, animal feed, personal care items, and food supplements. The most important advantages of microalgae biomass over traditional raw materials are as follows:

- They are usually produced 10–100 times more than ground crops;
- They capture carbon at high yields;
- They contain lipids or carbohydrates at high concentrations that may be useful for biodiesel or bioethanol creation;
- They may be cultivated in seawater, brackish waters, or even sewage;
- They offer the possibility of utilizing non-arable land.

Microalgae biomass can form a continuous biofuel production chain, similar to traditional oil refineries, because it can be harvested all year. Furthermore, on-site processing of the biomass generated and microalgae cultivation facilitate integrated and sequential production of multiple products while lowering logistical and administrative expenses at biorefinery facilities. Nevertheless, producing commercially viable biofuels from microalgae presents substantial technological obstacles, including the following:

- Development of adapted microalgae strains at high biomass yields;
- Effective methods of cultivation and crop protection;
- Advancement of harvesting and lipid production methods;
- Optimization of biofuel conversion and production processes and other byproducts [52].

Value-added products like astaxanthin and beta-carotene are already produced by microalgae. To produce specialized products, microalgae like *Spirulina platensis*, *Haematococcus pluvialis*, *Chlorella vulgaris*, and *Dunaliella salina* are cultivated. One particular product, such as astaxanthin or beta-carotene, is produced, and the other important components, such as proteins and carbohydrates, are wasted or remain unutilized. The overall benefits of microalgae cultures could be greatly increased if multi-product production were achieved. In this way, biomass would be fractionated and exploited, and this would lead to economic viability, a method that is also applied in classic refineries of petrochemical or petroleum products. This type of concept is called a multi-product biorefinery, and it is a production scheme that does not aim at a single product but at the full utilization of all biomass fractions. Such a typical installation initially includes the stages of harvesting, cell wall rupture, and extraction of products. This is followed by further fractionation, such as the isolation of carbohydrates from water-soluble proteins [53] (Figure 3).

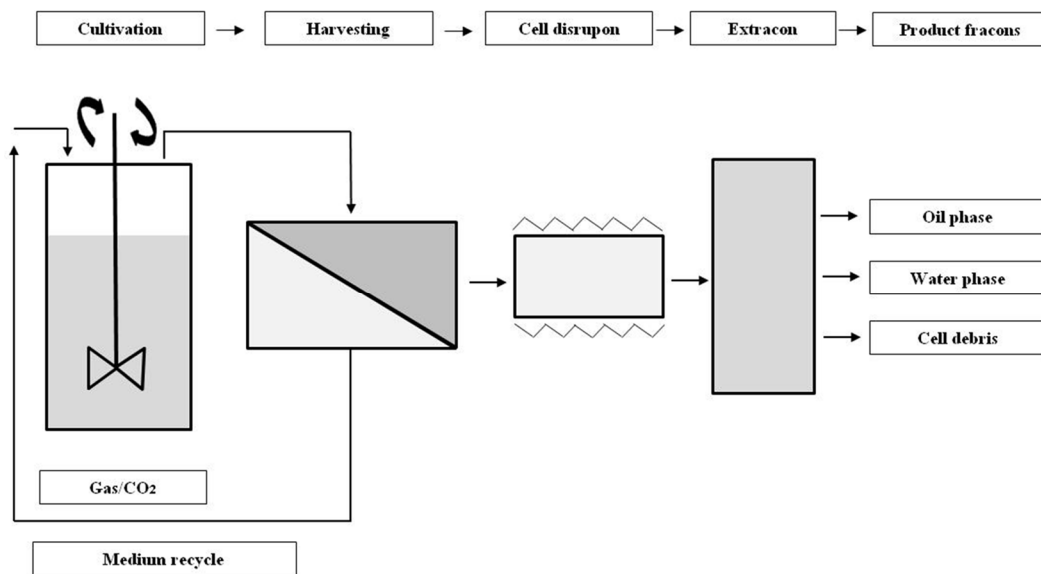


Figure 3. Simplified flow diagram of a multi-product microalgal biorefinery (modified from 't Lam et al. [53]).

In the concept of the microalgae biorefinery, there are advantages that integrate it into the principles of the circular economy (a production scheme where renewable materials are used as raw materials for the production of fuels and chemicals, and all byproducts are fully degradable). The biorefinery approach offers a solution to profitably exploit the full potential of microalgae through the recovery and separation of biomass components, as well as by minimizing waste generation. In this light, the microalgae biorefinery not only comprises sequential steps for separation of biomass fractions, but also a succession of functional units that groups laboratory-level research with culture development and subsequent processing stages, with little to no waste.

Laboratory-scale research involves the selection of the specific strain of microalgae based on the desired product to be produced and the sources and availability of low-cost nutrients for microalgae culture. The cultivation stage concerns the construction, operation, and optimization of a cultivation system with the following characteristics:

- ✓ It best matches the requirements of the selected microalgae strain in terms of mixing with the culture medium and light penetration;
- ✓ It reduces investment and operating costs;
- ✓ It offers sufficient biomass concentration and productivity for efficient post-processing.

Subsequent processing steps include the following:

- Biomass harvesting;
- Extraction of intracellular components through cell lysis;
- Fractional separation of ingredients to maximize commercial value products that can be obtained [54].

The microalgae biorefinery has so far not been a profitable activity. In commercial biotechnology products, the cost of post-processing amounts to 20–40% of the total production cost, while in multi-product microalgae biorefineries (a biorefinery production scenario that aims not only at one product, but at full utilization of all biomass components), it is significantly higher and amounts to 50–60%. This difference is mainly due to two main factors:

- Low biomass concentration in microalgae cultures (<3 g/L);
- The difficulty in applying mild techniques that provide the different desired biomass fractions without destroying other fragile substances present in biomass, to optimize and recover all valuable components (proteins, carbohydrates, and lipids) [53].

Disadvantages and difficulties exist at all the above stages.

- At the laboratory level, it is the selection of the appropriate strains, their engineering, and the nutritional approach that leads to the overaccumulation of the desired product;
- At the cultivation stage, and especially in industrial photobioreactors, there are high costs and low concentrations of biomass, reaching about 3 g/L, while reactors should process biomasses with at least ten times the concentration for their efficient operation and utilization;
- Finally, in later processes, designs are made for a single main product and the rest is often “waste” that needs to be disposed of, which adds to the cost [54].

Notable improvements, which enhance the concept of the biorefinery, have been made to the overall picture of the microalgae technology sector. The development of technology in cultivation systems (open and closed) has led to thinner crops, where light penetration was improved, resulting in increased photosynthesis and, consequently, an increase in biomass produced. These cultivation systems offer satisfactory concentrations of biomass produced, thus facilitating the processes that follow (harvesting, cell lysis, and product fractionation).

The dominant microalgae biorefinery optimization approaches are developed at two levels:

- At the first level, the strategy aims to maximize the production of the desired value-added product and the utilization of byproducts, through overaccumulation achieved in closed flat thin-surface photobioreactors. This includes the cultivation of *Haematococcus* sp. for astaxanthin production where high-yielding costly technology is used to optimize production. The remaining biomass after astaxanthin extraction contains mainly proteins, carbohydrates, and pigment residues. Neither the yields or quality of these substances correspond to the market for food additives, but they can be used in biofuel production or for feeding animals.
- The second level of the approach aims at the production of multiple products in successive stages. The slow part of the exponential phase of crop growth is selected, where nitrogen depletion occurs and exploitable products accumulate. The priority of extraction of products is regulated by the fragility of substances (soluble proteins, dyes, lipids, and starch). In this case, robust and dynamic strains such as *Scenedesmus*, *Chlorella*, *Tetraselmis*, *Phaeodactylum*, and *Nannochloropsis* spp. are used, resulting in a wide range of substances from medium value (PUFAs, functional proteins) to low value (starch or insoluble proteins). With a chain pattern of extractions, the water-soluble components (mainly proteins and hydrocarbons) are obtained first by gentle cell wall disruption and centrifugation. This is followed by lipid extraction and, in some cases, the separation and enrichment of specific lipids such as PUFA. The last stage examines the recovery and utilization of the starch fraction in the residue, for application in the production of bioplastics.

In conclusion, it is proven that the maximum utilization of all components of microalgae biomass, such as water-soluble proteins that can be food additives instead of being used for animal feed, is a very important point of research direction. In addition, other substances may be present in the residual fraction after extraction of the main compounds, which need to be investigated for their possibility of producing new commercial products. Finally, the costs of developing and introducing complex and costly technologies for further recovery of biomass fractions should guarantee reciprocal economic benefits [54].

As the biggest challenge in the industrial production of biofuels from microalgae is the high cost, it is extremely important to use waste as a growing medium to achieve a reduction in biomass production costs. Moreover, the production of a single biofuel from microalgae is not economically viable. It is therefore imperative to link biofuel production with other activities for the energy balance. The use of waste or byproducts, where one form of energy is produced from the production of another form of energy, is an excellent way to achieve economic viability. An integrated biorefinery approach to maximize the utilization of microalgae biomass and the use of waste for microalgae cultivation as a growth medium is a viable and cost-effective solution for a greener future [55]. Essentially, integrating microbiomass production with waste treatment leads to environmental sustainability, as

well as to improved economic performance. The main products are recycled water (purified waste water), biomass-derived products such as biofuels, and soil improvers as residues of the biofuel production process. The two main approaches in this direction are as follows:

- The use of microalgae for the direct treatment of waste, with the effluents then disposed of for off-site use (i.e., the waste is used only once to produce the microalgae biomass);
- The use of treated or raw wastewater as a growing medium for microalgae biomass production, with wastewater then recycled after treatment [56].

Since microalgae biomass production of biofuels is not yet at a sustainable level, integrating them into other production processes is an intriguing way to address both environmental and economic concerns. Within the framework of a biorefinery, microalgae biomass can be produced using raw materials supplied by related industrial units, or it can be utilized for the recovery of its constituent parts. The integration of all processes and the use of industrial waste from various sources for microalgae cultivation have been regarded as crucial strategic decisions. The process of converting raw materials into final products includes the production of liquid and gaseous waste from industrial plants. Typically, in these situations, the waste streams go through a number of technical processes before being released back into the environment. The application of heat and mass integration strategies with other production processes is an alternative to conventional waste treatment and offers a genuine chance for appropriate, affordable waste management. In Psachoulia et al. [57], treated wastewater was used as a dilution medium, which helped to improve the environmental sustainability of the water management strategy without affecting the rate of culture growth.

Certain kinds of liquid and gaseous waste streams contain both organic and inorganic content in addition to CO₂, making them appropriate for use as carbon and other nutrient sources when growing microalgae. Combining the use of microalgae with industrial facilities' processes has the following immediate benefits:

- Reduction of the amount of energy, steam, and water needed;
- A decrease in the amount of effluent used for waste remediation;
- A decrease in the quantity of pollutants that eventually reach the environment.

In addition, the thermal and electrical energy provided by the installed units is exploited through the microalgae.

The integration of microalgae biomass production with a production activity can be applied in the case of sugar and biofuel factories, as in this case both energy such as steam and electricity, and necessary substances such as coal, minerals, and water, are offered. The sugar and energy sector has been particularly dynamic in Brazil since the 1970s, while the country is developing one of the most successful large-scale biofuel production programs in the world. This sector combines the above characteristics with additional advantages such as low land value and increased sunshine. Brazil's average yearly PAR, or photosynthetically active radiation, falls within the 400–700 nm wavelength range. The ability of photosynthetic organisms to grow crops depends critically on the availability of this unique form of radiation. Together with sugar and biofuel factories, all of these elements help to create a favorable environment in the nation for the growth of microalgae crops. Additionally, since the cost of raw materials has a significant impact on the production of bioethanol, the establishment of a biorefinery concept between the production of bioethanol and concurrent microalgae biomass production adds investment and environmental benefits to the project's overall economic viability. Sugar cane processing in Brazil is currently carried out through three different types of production schemes:

- Sugar factories, which produce only sugar;
- Stand-alone bioethanol plants producing hydrated or anhydrous ethanol;
- Sugar factories connected to bioethanol production plants producing sugar and bioethanol.

Integration between sugar factories and other industrial activities is already taking place in Brazil in specific cases. Depending on the type of production process, integrated facilities benefit from integrated management of the supply of raw materials and other

auxiliary materials, intermediate production flows of products, finished products, or from surplus energy produced in sugar factories. To the above are added the administrative and research activities and facilities, as well as the equipment and maintenance capacity of the facilities. The possibility of annexing microalgae biomass production to existing sugar mills is currently being intensively examined in the industrial environment and in the scientific community. In the unified sugarcane–microalgae biorefinery scheme, biodiesel can be produced from microalgae biomass, which will be used as fuel for the needs of sugar cane cultivation. In this way, the total greenhouse gas emissions associated with bioethanol production are reduced by approximately 30%, as the microalgae for their growth capture 50% of the CO₂ produced in the fermentation process of ethanol production. This approach has been attempted in Brazil since 2014, at the level of factory production, to produce 100 thousand tons of biodiesel in a unified sugar industry production scheme with microalgae biomass production. The main available sources of the sugar-biofuel industry that could directly contribute to microalgae cultivation are the following:

- ✓ CO₂ produced during the fermentation process for ethanol production or produced during the operation of boilers producing heat and energy for the needs of the sugar mill. This ensures the supply of CO₂ for the cultivation of microalgae.
- ✓ The byproduct of bioethanol production after fermentation is vinasse, which can be used as a culture medium for the development of heterotrophic or mixotrophic crops.
- ✓ The surplus electricity produced in the sugar cane plant can be used directly in the various stages of microalgae growth and processing.

In case that the production scheme is second-generation, carbohydrates produced from microalgae biomass can be fermented either together with sugar cane juice and molasses, by fermenting microorganisms that can ferment pentoses and hexoses, or in independent fermentors [58]. Vinasse is an acidic, dark brown liquid, rich in organic compounds (e.g., glycerol, lactic acid, sugars), nitrogen, phosphorus, and ions, such as potassium, calcium, and magnesium, and produced in huge quantities. For every liter of ethanol produced from sugar cane, 12–14 L of vinasse is produced as a byproduct after fermentation. Its main use is as a fertilizer on sugarcane plantations, but it has the disadvantages of altering soil composition and reducing sugar cane productivity when applied continuously [59]. For the production of first-generation ethanol, the production process includes purification and cutting of sugar cane, extraction to produce dilute juice, concentration into concentrated juice, juice purification, fermentation for ethanol production, and then distillation and dehumidification of the produced bioethanol. The residue of the extraction is bagasse (lignocellulosic synthesis), of which 50% is used for the production of second-generation ethanol, and the rest as fuel in the combined heat and power plant. For the production of second-generation ethanol, bagasse is subjected to physicochemical treatment to break the lignin–cellulose–hemicellulose bonds and increase the percentage of cellulose for the enzymatic hydrolysis that follows. Through hydrolysis, cellulose is converted into monomer sugars, and the resulting sucrose solution is concentrated. In the next step, it is mixed with the dense juice from the stream of the first generation, and all together sent to fermentation for ethanol production. Finally, for the production of value-added products from microalgae biomass such as lipids and carotenoids, supercritical extraction is preferred, which is an effective method. It is a “clean” technology applied in the context of a holistic approach of the biorefinery, for the recovery of substances from plant materials, and is preferable to chemical treatments [60].

Furthermore, palm oil production has contributed significantly to the economy of several tropical countries, such as Malaysia, which is the world’s largest producer. Palm trees are important crops due to the high productivity and high oil yields from small areas of land. The palm oil production process produces high pollutant load (POME) waste, which is discharged into the environment after being treated, to reach the required discharge limits [61]. The use of microalgae for processing POMEs in combination with biogas installations in palm oil plants can simultaneously remediate waste and reduce the cost of microalgae biomass production given the high concentrations of nutrients in them.

In this direction, a microalgae biorefinery can be incorporated into the processing of POME, which will grow with the residue of anaerobic treatment, binding CO₂ and producing lipids and energy through combustion, after harvesting and dehumidification [62]. Biomass is produced, which then, through solvent extraction of hexane and isopropanol, produces lipids, and by the transesterification process, biodiesel and glycerol are produced, while the residue is used for combustion for energy production. The results show that from a techno-economic point of view, this processing route produces the highest profit, which amounts to 7.73×10.5 USD/year [62].

For the development of sustainable biofuel production processes, the selection of an appropriate extraction solvent is still a major obstacle, despite the fact that many different approaches have been reported. Lipid extraction uses conventional organic solvents, which are inexpensive and simple to use, and are derived from petroleum products. They require more processing steps, are toxic, and consume large amounts of time. Furthermore, they react with other substances to worsen the product's quality, and the majority of them are flammable. European directives such as REACH (2006/1907/EC) tightly control their use. The use of "green" solvents (green extraction technologies) is being researched to improve the properties of energy requirement, ecological friendliness, non-toxicity, and efficient lipid extraction in order to address the aforementioned challenges. Terpenes, for instance, are naturally occurring green solvents with exceptional chemical and technical qualities, and are extracted from citrus fruits and many other plants. With no change in the composition of fatty acid methyl esters (FAMES), terpene application produced a higher yield in the biomass extraction of *Chlorella vulgaris* when compared to hexane extraction. Recyclable, green solvents function as a solvent system that eschews recovery—a step that is typically taken in conventional processes. Therefore, in a variety of industrial extraction processes, these solvents may be used in place of conventional and other environmentally friendly solvents. This reduces post-processing, saves energy and time, and produces fewer or no byproducts to easily offset the high cost of green solvents. The economic feasibility of biofuels produced from microalgae could be enhanced through multi-product cogeneration. For instance, *Chlorella* sp. and *Tetraselmis suecica* hold great promise for clean and sustainable CO₂ capture mechanisms in the future. They can also be used as feedstock for the anaerobic fermentation of biomass by the bacterium *Clostridium saccharoperbutylaticum*, which will produce acetic and butyric acid, as well as biofuels (ethanol, butanol, and acetone). Since additional extraction and separation processes could result in excessive costs for the biorefinery, virtually every biorefinery scheme needs to be evaluated separately to ensure that the highest level of sustainability is achieved both economically and environmentally. This evaluation should primarily focus on the choice of technologies, raw materials, and products [34]. According to Kokkinos et al. [63], *Chlorella vulgaris* is the most preferable strain used for nano-catalytic conversion into biofuel, due to its high lipid content. In terms of waste, the creation of the perfect integrated system that combines the anaerobic digestion of waste with the cultivation of microalgae is thought to be essential for the achievement of microalgae biomass production at a reasonable cost. The required perspective in this regard is offered by an integrated biorefinery that uses anaerobic digestion to treat activated sludge and uses anaerobic waste for microalgae cultivation. An efficient way to lower the overall carbon footprint of the sludge disposal process is to combine anaerobic sludge digestion with microalgae culture, which is a promising and reasonably priced method. Effluents from anaerobic processes have elevated COD, and high levels of total nitrogen (TN), total phosphorus (TP), and volatile fatty acids (VFAs). Because the conversion of organic carbon is not complete, the production of biogas (CH₄ or H₂) is relatively low and of lower purity. While the CO₂ in biogas can serve as an inorganic carbon source for a mixotrophic microalgae culture, the VFAs that are readily metabolized by microalgae and organic matter as COD in the effluent act as an organic carbon source for the microalgae. Biofuels can be produced from lipid-rich microalgae biomass, and the residual biomass and sludge can be treated thermochemically to produce biofuels or biochar, which will enhance the biorefinery system. The residual biomass can be removed through lipid extraction.

Pyrolysis can be used for the digested sludge and microalgae residue to produce biochar, which can be used as an adsorbent to remove contaminants or as a soil conditioner. Liquid or gaseous biofuels are produced by recovering the energy from the microalgae residue through a mild process known as pyrolysis, or flaking [64].

8. Microalgae in Environmental Biomonitoring

Microalgae are considered to be very useful in the treatment of urban wastewaters since they can be utilized as pre- or post-treatments [65,66]. Co-culture of microalgae positively affected the biodegrading microbes and showed greater treatment effectiveness [67]. Despite the advantages of using biological systems to treat urban wastewater, their effectiveness is evaluated by using chemical and physicochemical parameters. These effluents are made out of an incredible assortment of ecological pollutants, which, on multiple occasions, are just inadequately or not removed. Additionally, these pollutants may, contingent upon their properties, have added substance, synergistic, or hostile actions with harmful consequences for natural life and people [68]. Apart from the rubbish that is dumped into rivers by people and businesses, a lot of pesticides are also known to enter the aquatic ecosystem through agricultural runoff or leaching [69]. Aquatic ecosystems may quickly find some of these organic contaminants.

Government associations have carried out physical and synthetic estimations that offer quantitative information on the degrees of water contamination and debasement to address their interests about keeping up with the nature of oceanic assets. However, since test organisms respond to all of the compounds in wastewaters, traditional physicochemical analyses ought to be supplemented by bioassays utilizing various species [70]. In this manner, a compelling and productive approach to deciding the harmfulness of different pollutants and microorganisms implanted in the environment is viewed as vital. The biomonitoring through water ecotoxicology tests can likewise give significant pieces of knowledge about the possibly poisonous impacts of treated wastewaters.

Microalgae play a major role in the first level of the food chain, so they can be an effective tool in this situation as biological markers of pollution in ecotoxicity [71]. Because of their ability to absorb heavy metals and harmful pollutants from wastewater, microalgae are thought to be a helpful indicator of the health of the environment [72]. Furthermore, microalgae can function as representative phytoplankton in the water, and can be used to indicate the trophic level and status of the water quality [73]. Increases in phosphorus and nitrogen in water systems result in increased biological productivity, which in turn produces phytoplankton. High concentrations of nutrients, such as phosphorus and nitrogen, have been shown to increase phytoplankton biomass, which is a sign of a poorly functioning waste system. Furthermore, in comparison to other microorganisms, microalgae species exhibit a greater degree of sensitivity when exposed to specific groups of hazardous contaminants [74]. Consequently, the present article highlights the importance of conducting biomonitoring by wastewater treatment plants as an alternative to cleaner technology.

9. Conclusions

Microalgae represent a biosystem of high environmental performance. In fact, microalgae culture can be applied to effectively treat a wide range of effluents, and bioprocessing appears to be a viable option.

Moreover, along with the capability of microalgae crops to be cultivated in low-cost wastewater, their additional properties to capture important CO₂ greenhouse gas emissions for their photosynthetic growth may contribute towards mitigation of present climate change effects.

Concurrent with wastewater bioremediation, microalgae present a feasible and encouraging natural resource for generating value-added compounds, in the context of human and environmental health promotion. Antioxidants and dietary supplements beneficial to human health, medications, cosmetics, human nutrition products, animal and aquatic feed, soil conditioners, and others, are included. Furthermore, high-value substances

such as astaxanthin, β -carotene, DHA, EPA, and fatty acid methyl esters can be produced. The recovery of valuable ingredients from microalgal biomass and their valorization into beneficial products is also considered to provide an effective option for reducing the cost for simultaneous large-scale production of third-generation biofuel, including biodiesel, bioethanol, biomethane, and biohydrogen. The optimization of biofuel development concurrently with the creation of value-added products passes through the modern biorefinery concept. The sustainability of microalgal biorefineries, along with the economic and environmental aspects, remains a critical objective.

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